定时器&PWM应用编程

1.使用STM32F103的 Tim2~Tim5其一定时器的某一个通道pin（与GPIOx管脚复用），连接一个LED，用定时器计数方式，控制LED以2s的频率周期性地亮-灭。

2.接上，采用定时器PWM模式，让 LED 以呼吸灯方式渐亮渐灭，周期为1~2秒，自己调整占空比变化到一个满意效果；使用Keil虚拟示波器，观察 PWM输出波形。

一、实验原理

1. STM32通用定时器概述

STM32F103C8T6属于中等容量产品，内置4个通用定时器（TIM2~TIM5）。每个通用定时器包含一个16位自动装载计数器，由可编程预分频器驱动，具有4个独立通道，可用于输入捕获、输出比较、PWM生成和单脉冲模式输出。

通用定时器的时钟源来自APB1总线。当APB1预分频系数为1时，定时器时钟等于APB1时钟频率；否则定时器时钟为APB1时钟的2倍。在本实验的72MHz系统时钟配置下，APB1总线频率为36MHz，但由于APB1预分频系数不为1，定时器时钟为72MHz。

2 .定时器中断工作原理

定时器中断的触发流程如下：

1. 预分频器（PSC）对定时器时钟进行分频，得到计数器的实际工作时钟CK_CNT
2. 计数器（CNT）在每个CK_CNT周期递增（或递减）计数
3. 当CNT值达到自动重装载寄存器（ARR）的设定值时，产生更新事件（UEV）
4. 硬件置位状态寄存器（TIMx_SR）中的更新中断标志位（UIF）
5. 如果中断使能寄存器（TIMx_DIER）中的UIE位为1，则向NVIC发出中断请求
6. CPU响应中断，跳转到对应的中断服务函数（TIMx_IRQHandler）执行
7. 在中断服务函数中必须手动清除UIF标志位
   定时周期的计算公式为：

$$T_{out} = \frac{(ARR + 1) \times (PSC + 1)}{f_{clk}}$$

其中TIMx_CLK为定时器输入时钟频率，ARR为自动重装载值，PSC为预分频系数

3 .PWM工作原理

PWM（脉冲宽度调制）通过改变输出脉冲的占空比来控制输出信号的平均电压。占空比定义为：

占空比=高电平时间周期×100%占空比=周期高电平时间×100%

STM32的PWM输出基于定时器的比较功能实现。每个通道都有一个捕获/比较寄存器（CCRx），当计数器的值CNT与CCRx匹配时，输出电平发生翻转：

- **PWM模式1**：向上计数时，CNT < CCRx时输出有效电平，否则输出无效电平
- **PWM模式2**：向上计数时，CNT < CCRx时输出无效电平，否则输出有效电平

PWM周期由ARR决定，占空比由CCRx决定：

$$\text{占空比} = \frac{\text{ARR}+1-\text{CCRx}}{\text{ARR}+1} \times 100\%$$

例如，PSC=71，ARR=499时，PWM周期 = 500 × 72 / 72,000,000 = 0.5ms，频率为2kHz。

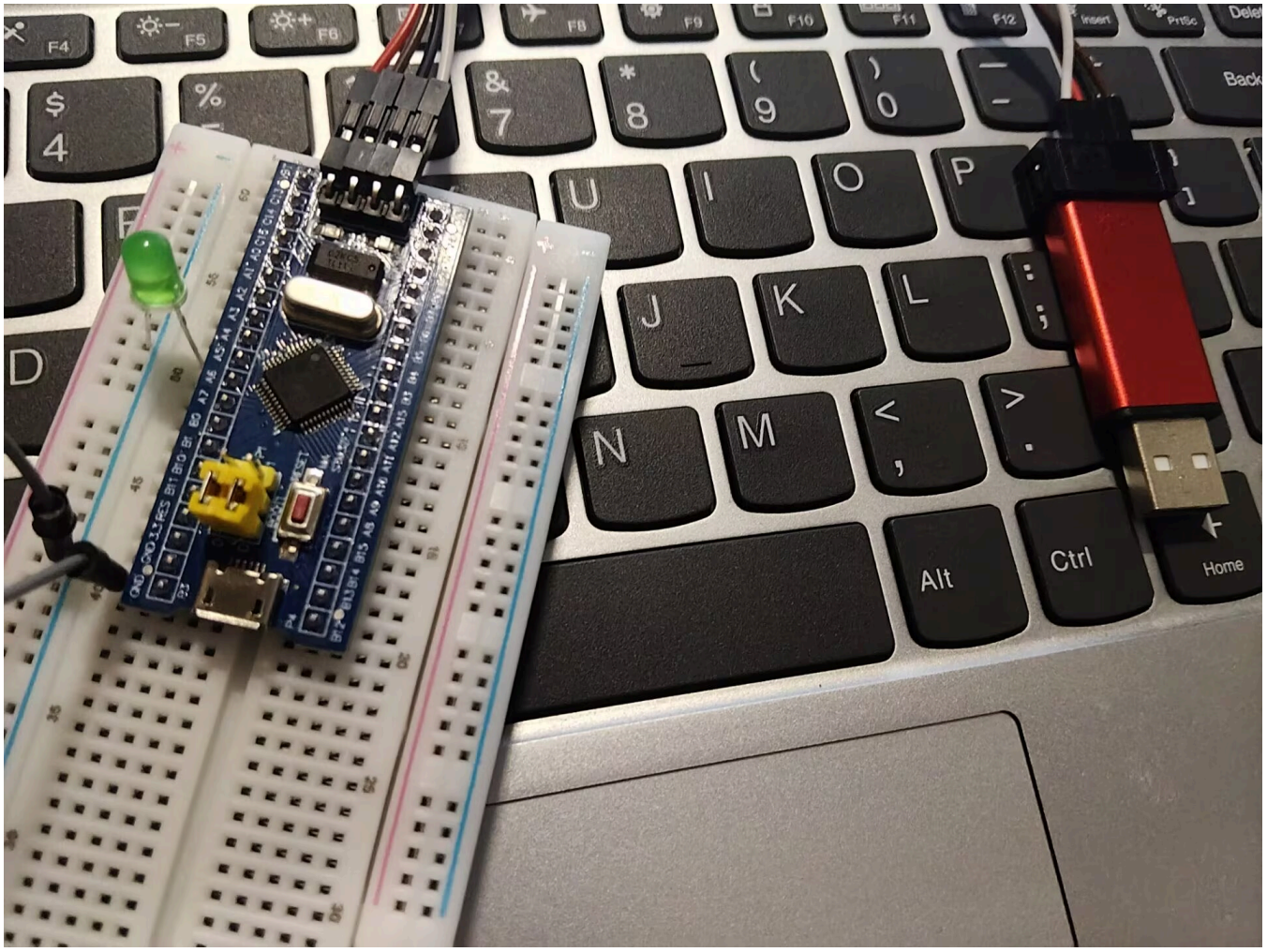
4 .引脚映射

本实验使用TIM3的通道1（PA6引脚）输出PWM信号驱动LED。STM32F103C8T6的TIM3通道映射如下：

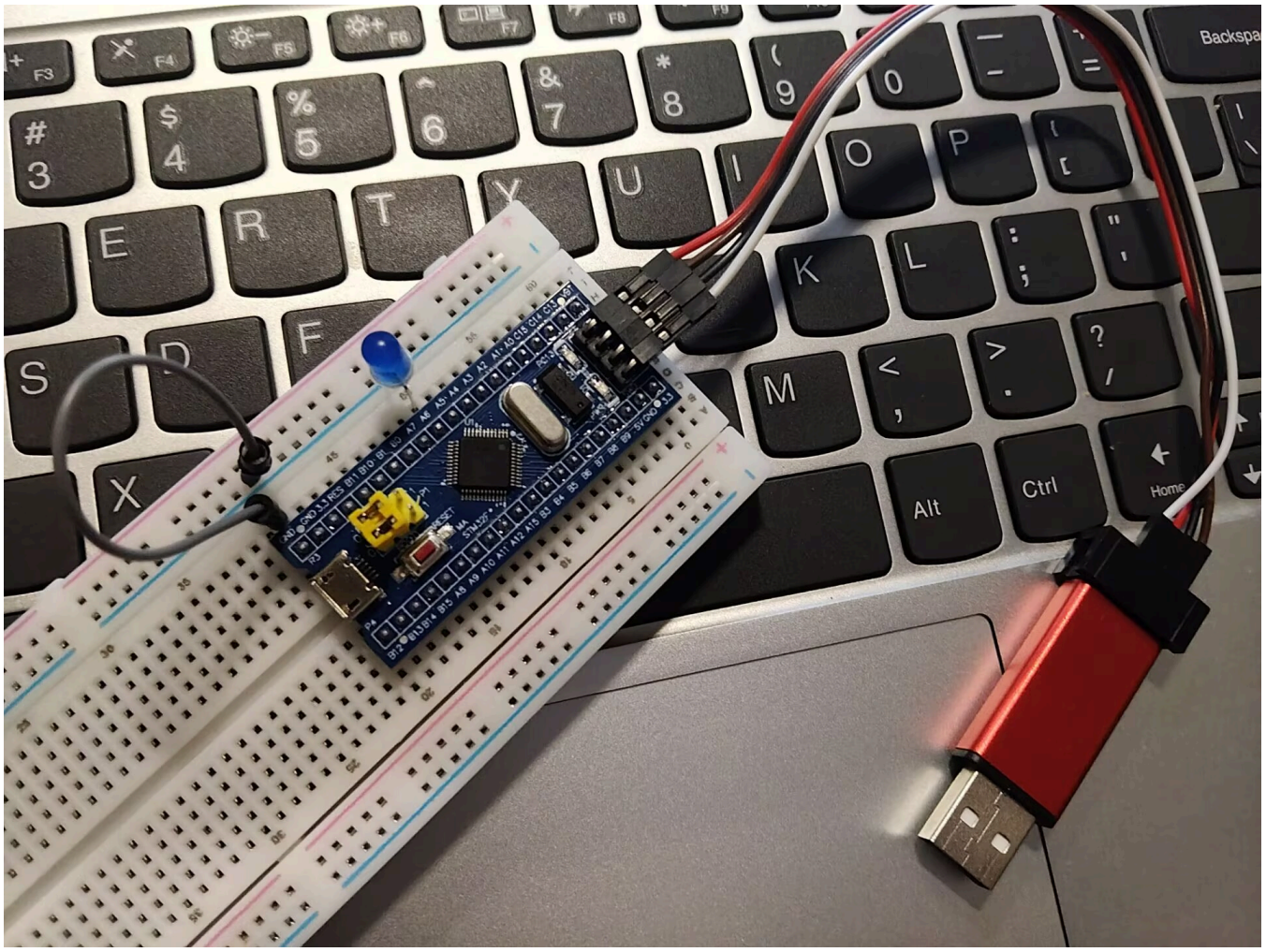
定时器通道	引脚	说明
TIM2_CH1	PA0	PWM输出
TIM2_CH2	PA1	PWM输出
TIM3_CH1	PA6	PWM输出（本实验使用）
TIM3_CH2	PA7	PWM输出
TIM3_CH3	PB0	PWM输出
TIM3_CH4	PB1	PWM输出

二、硬件连接

1.定时器

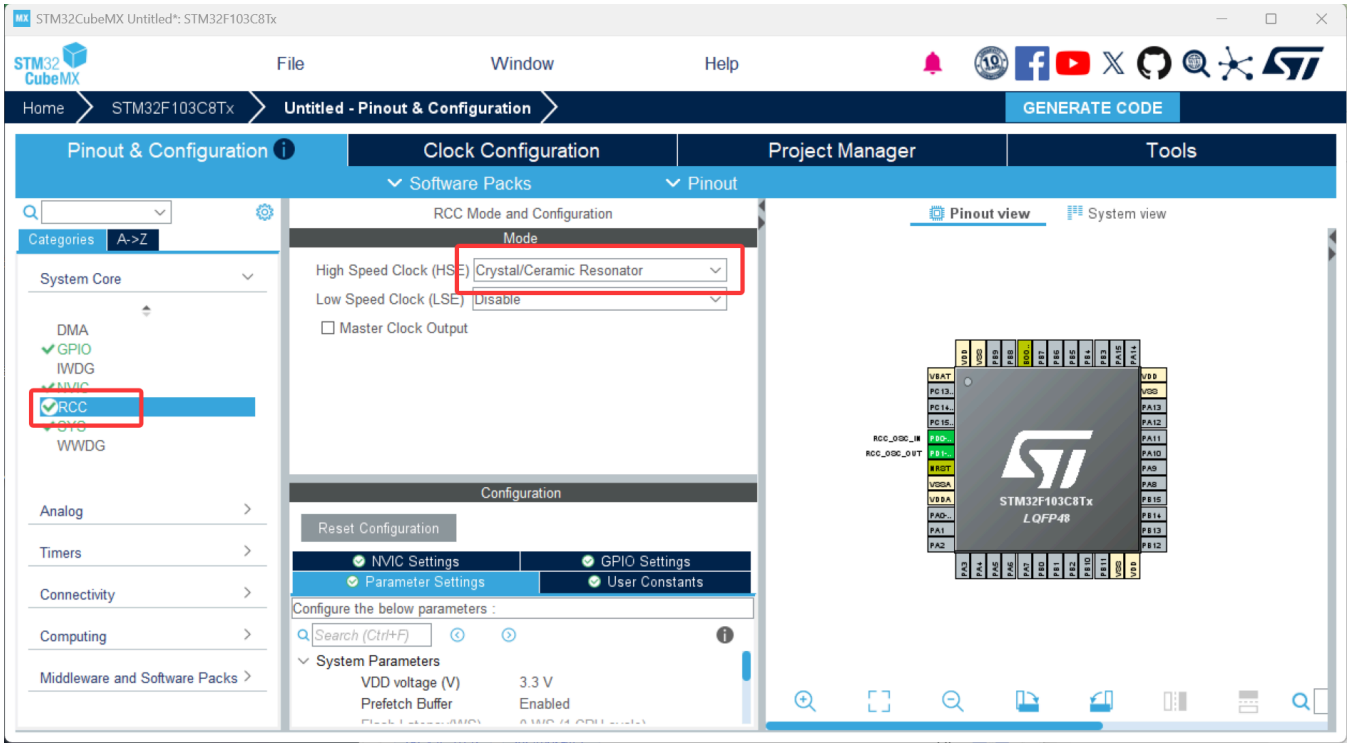


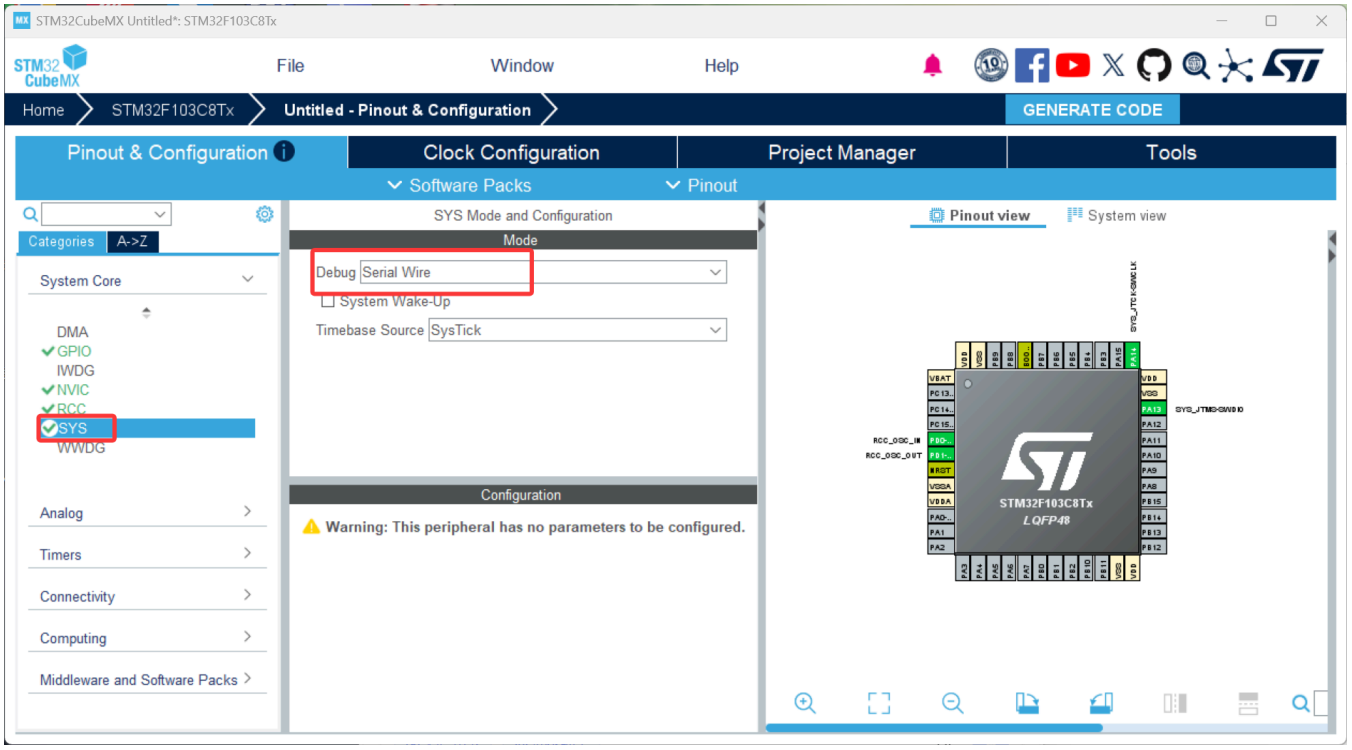
2.定时器PWM

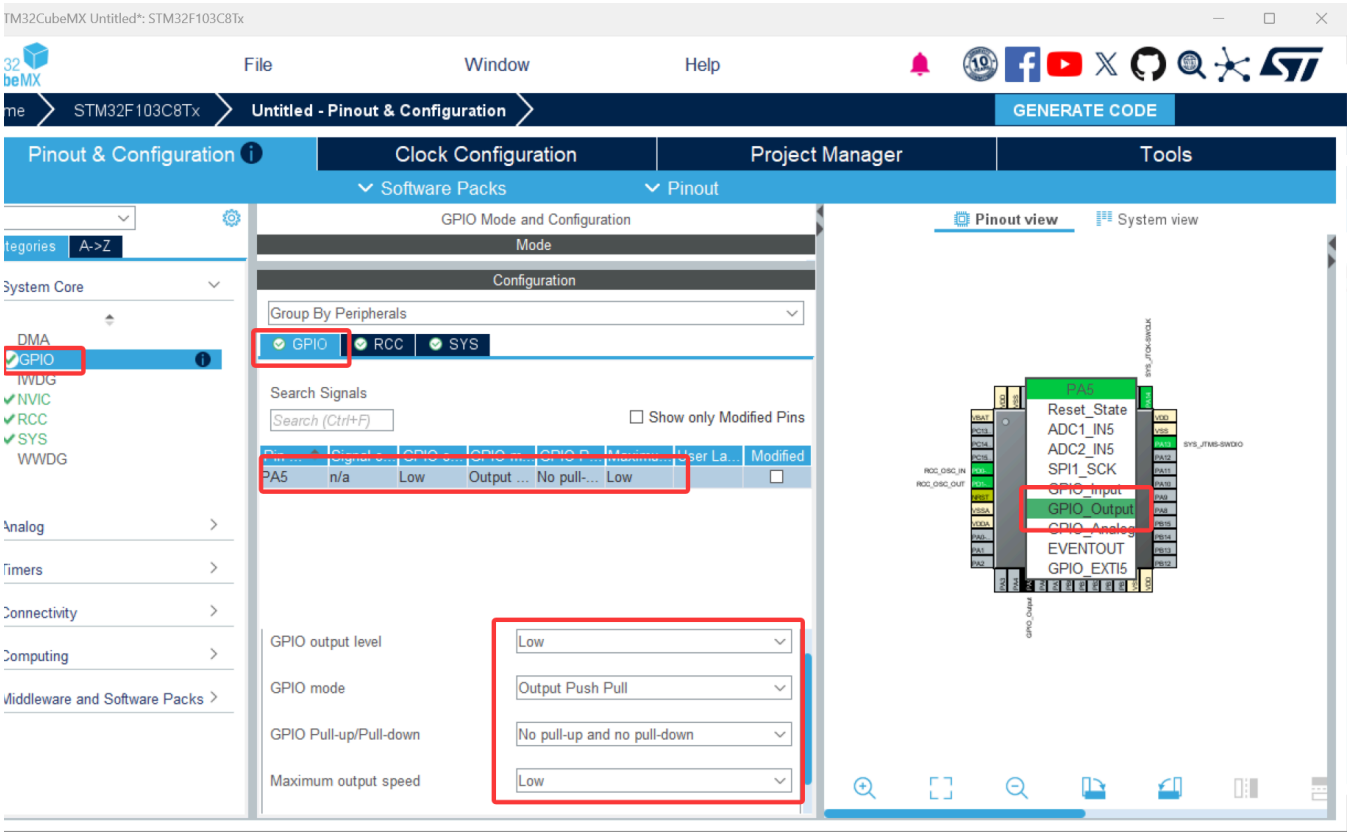


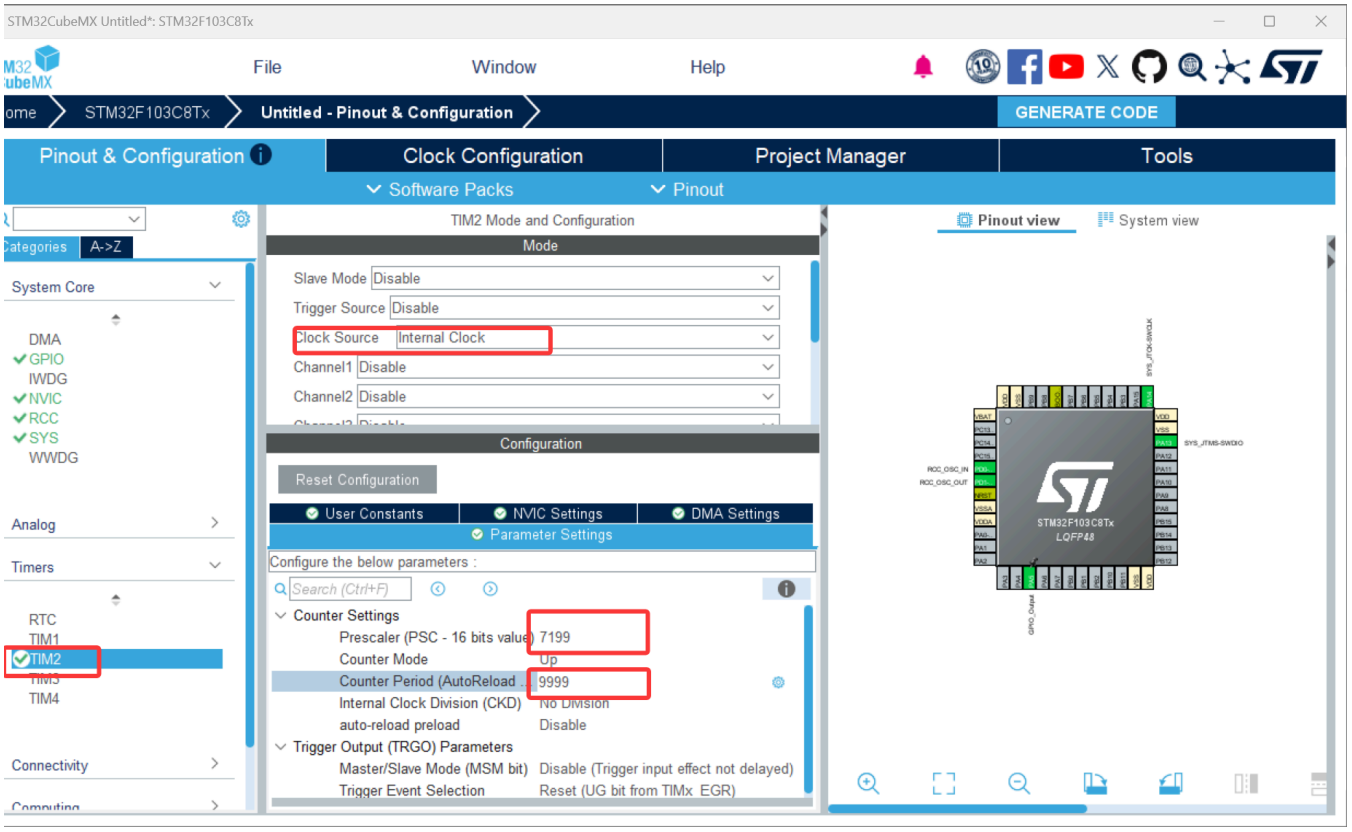
三、用定时器计数方式，控制LED以2s的频率周期性地亮-灭

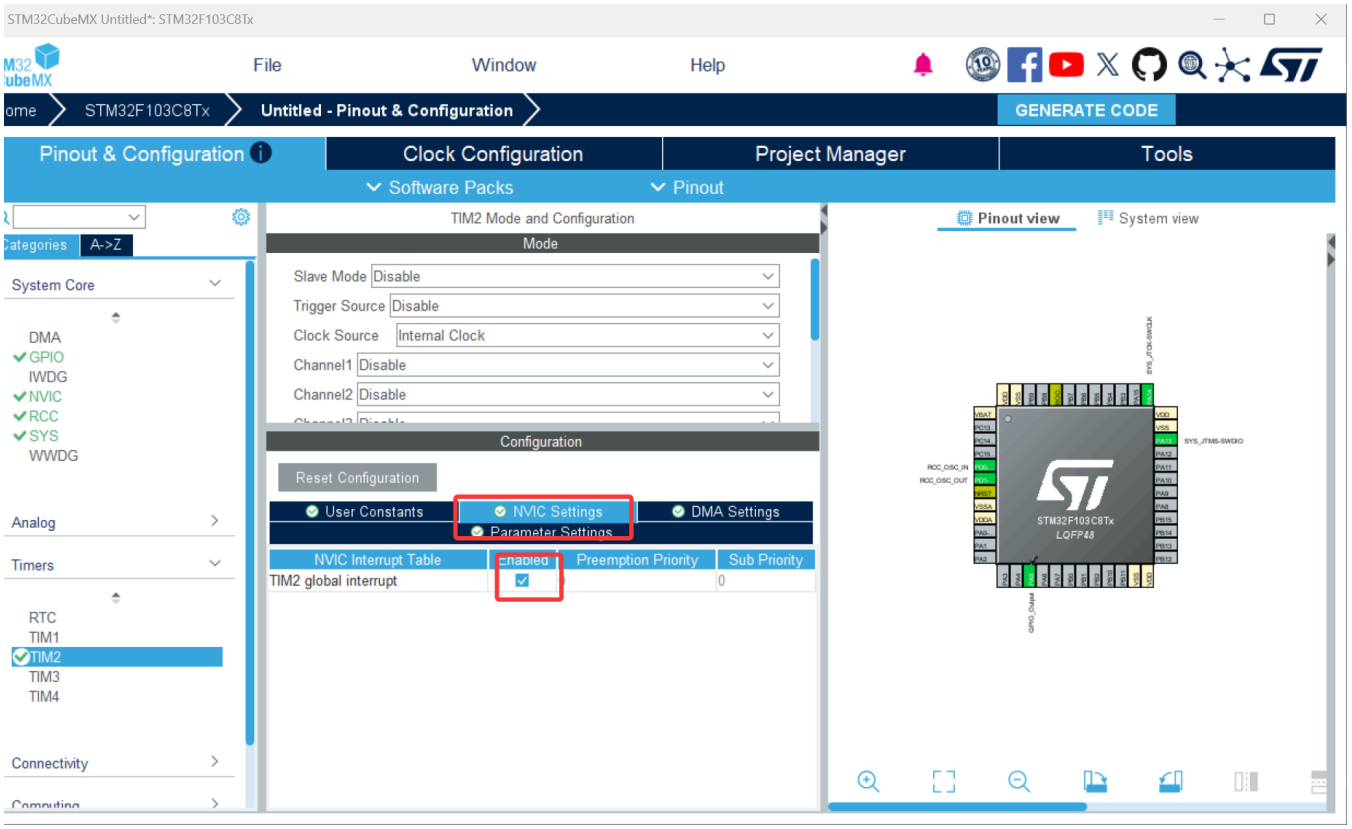
1.创建工程

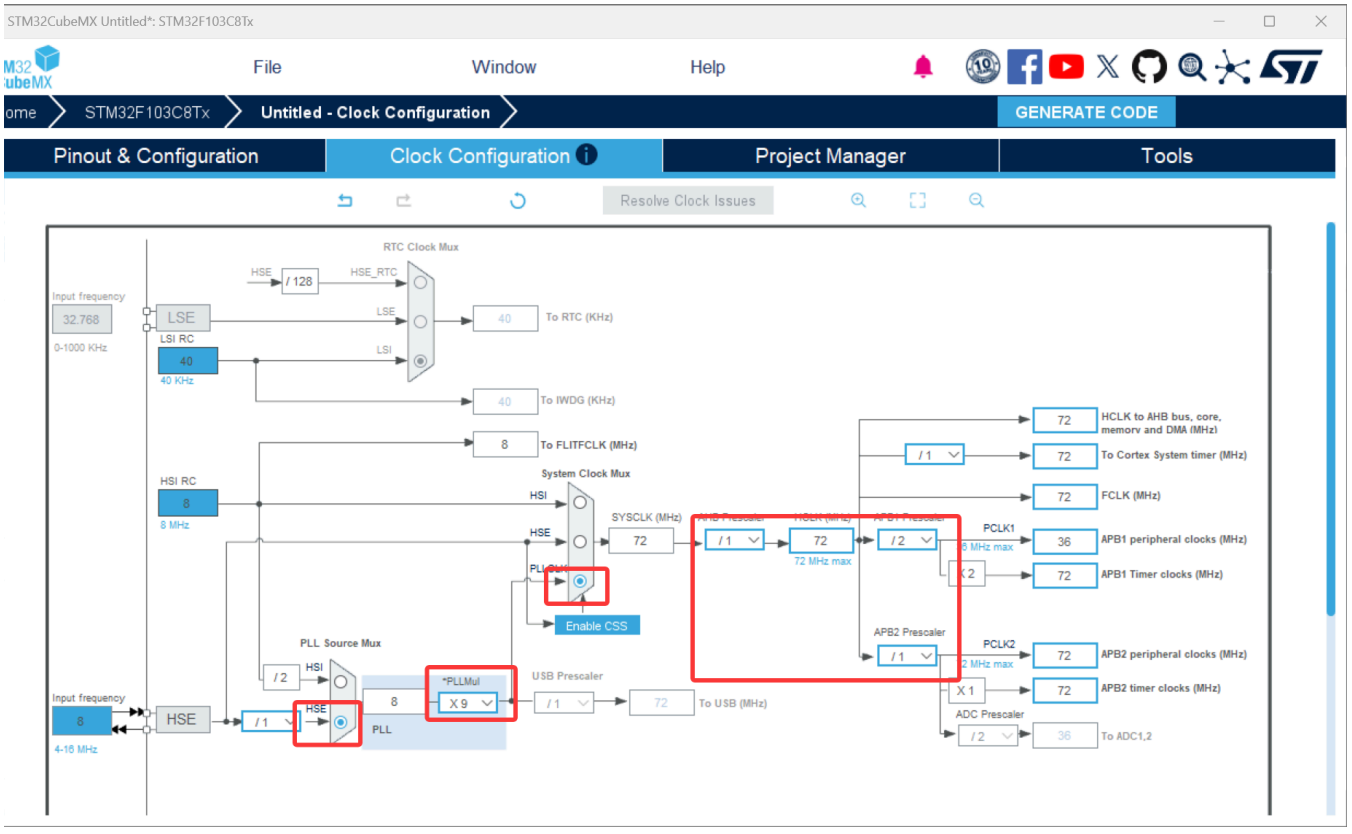

STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
Home
STM32F103C8Tx
Untitled - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Categories
A->Z
System Core
DMA
GPIO
IWDG
NVIC
RCC
SYS
WWDG
Analog
Timers
Connectivity
Computing
Middleware and Software Packs
RCC Mode and Configuration
Mode
High Speed Clock (HSE)
Crystal/Ceramic Resonator
Low Speed Clock (LSE)
Disable
Master Clock Output
Configuration
Reset Configuration
NVIC Settings
GPIO Settings
Parameter Settings
User Constants
Configure the below parameters :
Search (Ctrl+F)
System Parameters
VDD voltage (V)
3.3 V
Prefetch Buffer
Enabled
Pinout view
System view
STM32F103C8Tx
LQFP48

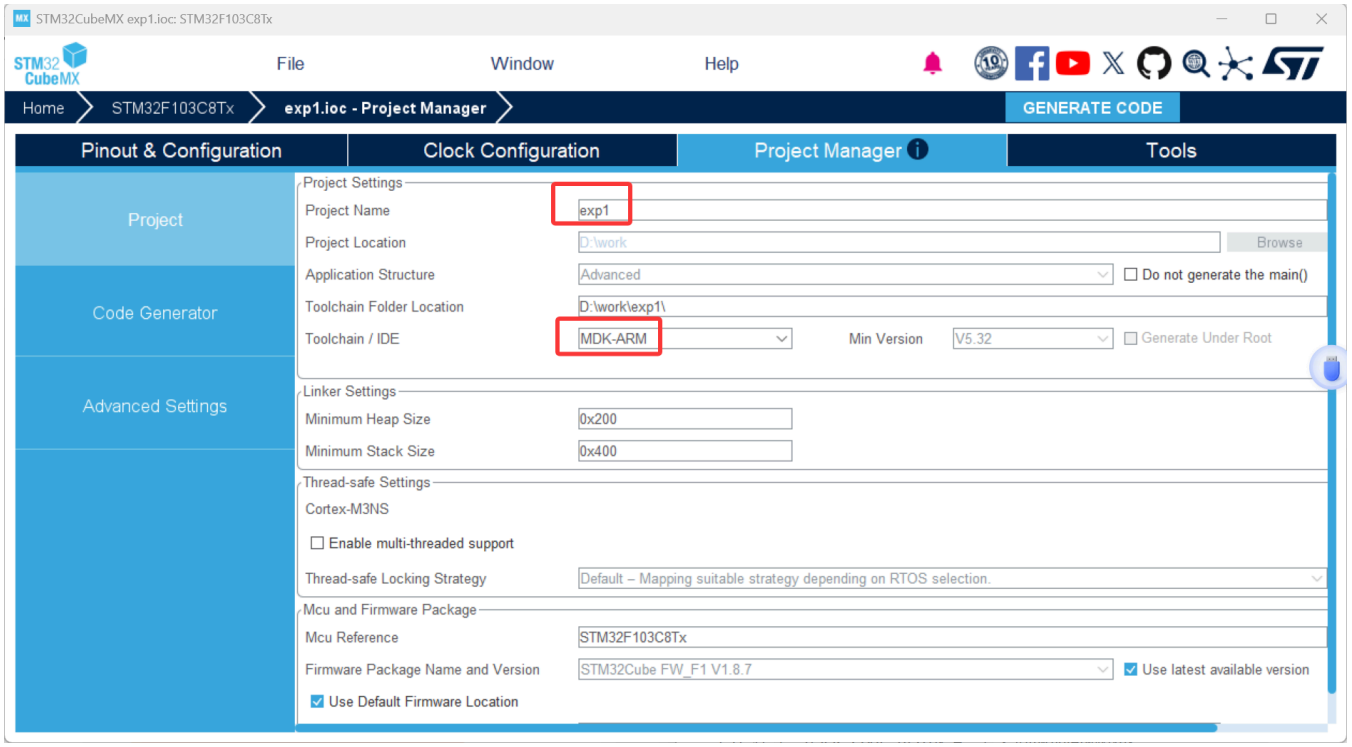

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SYS Mode and Configuration
Mode
Debug
Serial Wire
System Wake-Up
Timebase Source
SysTick
Configuration
Warning: This peripheral has no parameters to be configured.
Pinout view
System view
STM32F103C8Tx
LQFP48

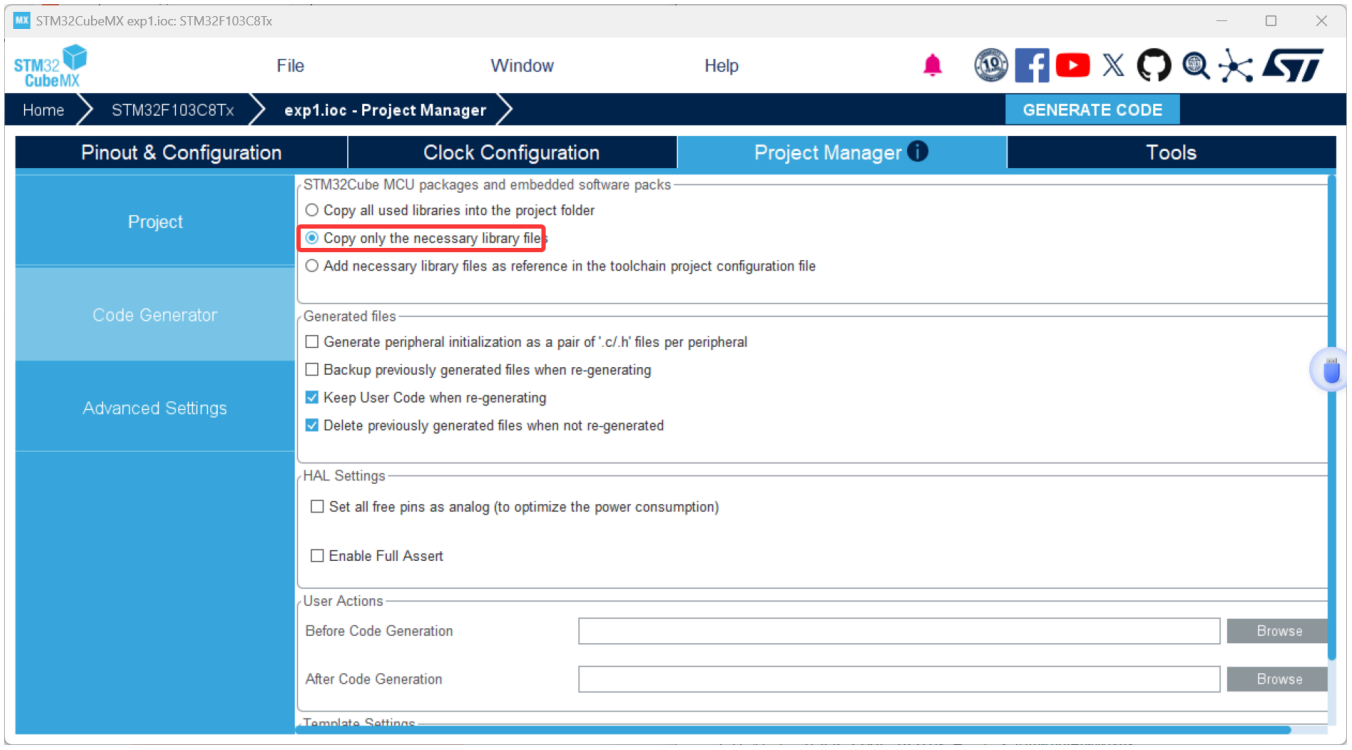

TM32CubeMX Untitled*: STM32F103C8Tx
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GPIO Mode and Configuration
Mode
Configuration
Group By Peripherals
GPIO
RCC
SYS
Search Signals
Search (Ctrl+F)
Show only Modified Pins
PA5
n/a
Low
Output ...
No pull-...
Low
Modified
GPIO output level
Low
GPIO mode
Output Push Pull
GPIO Pull-up/Pull-down
No pull-up and no pull-down
Maximum output speed
Low
Pinout view
System view
Reset_State
ADC1_IN5
ADC2_IN5
SPI1_SCK
GPIO_Input
GPIO_Output
GPIO_Analog
EVENTOUT
GPIO_EXTI5
System Core
DMA
GPIO
IWDG
NVIC
RCC
SYS
WWDG
Analog
Timers
Connectivity
Computing
Middleware and Software Packs


STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
STM32F103C8Tx
Untitled - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
TIM2 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Clock Source Internal Clock
Channel1 Disable
Channel2 Disable
Configuration
Reset Configuration
User Constants
NVIC Settings
DMA Settings
Parameter Settings
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value) 7199
Counter Mode Up
Counter Period (AutoReload ... 9999
Internal Clock Division (CKD) No Division
auto-reload preload Disable
Trigger Output (TRGO) Parameters
Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)
Trigger Event Selection Reset (UG bit from TIMx EGR)
Pinout view
System view
STM32F103C8Tx
LQFP48
System Core
DMA
GPIO
IWDG
NVIC
RCC
SYS
WWDG
Analog
Timers
RTC
TIM1
TIM2
TIM3
TIM4
Connectivity


STM32CubeMX Untitled*: STM32F103C8Tx
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STM32F103C8Tx
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Project Manager
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Pinout
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TIM2 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Clock Source Internal Clock
Channel1 Disable
Channel2 Disable
Configuration
Reset Configuration
User Constants
NVIC Settings
DMA Settings
Parameter Settings
NVIC Interrupt Table
Enabled
Preemption Priority
Sub Priority
TIM2 global interrupt
0
Pinout view
System view
STM32F103C8Tx
LQFP48


STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
STM32F103C8Tx
Untitled - Clock Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Resolve Clock Issues
RTC Clock Mux
Input frequency
32.768
0-1000 KHz
LSE
LSI RC
40
40 KHz
HSE
/ 128
HSE_RTC
LSE
LSI
To RTC (KHz)
To IWDG (KHz)
To FLITFCLK (MHz)
HSI RC
8 MHz
System Clock Mux
HSI
HSE
PLLCLK
SYSCLK (MHz)
Enable CSS
PLL Source Mux
/ 2
*PLLMul
X 9
PLL
USB Prescaler
To USB (MHz)
Input frequency
4-16 MHz
HCLK to AHB bus, core, memory and DMA (MHz)
To Cortex System timer (MHz)
FCLK (MHz)
PCLK1
APB1 peripheral clocks (MHz)
APB1 Timer clocks (MHz)
APB2 Prescaler
PCLK2
APB2 peripheral clocks (MHz)
APB2 timer clocks (MHz)
ADC Prescaler
To ADC1,2





2.代码撰写

(1) 在`/* USER CODE BEGIN PV */`之间添加:

```
uint8_t tim2_count = 0;  // 定时器中断计数变量
```

(2) 在`/* USER CODE BEGIN 2 */`之间添加:

```
HAL_TIM_Base_Start_IT(&htim2);  // 启动TIM2定时器并使能中断
```

(3) 在`/* USER CODE BEGIN 4 */`之间添加回调函数:

```c
/* 定时器更新中断回调函数 */
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
  if (htim->Instance == TIM2)
  {
    tim2_count++;
    if (tim2_count >= 2)  // 累计2次中断（即2秒）
    {
      tim2_count = 0;
      HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);  // 翻转PA5引脚
    }
  }
}
```

(4) 完整代码

```c
/* USER CODE BEGIN Header */
/**
  ******************************************************************************
  * @file           : main.c
  * @brief          : Main program body
  ******************************************************************************
  * @attention
  *
  * Copyright (c) 2026 STMicroelectronics.
  * All rights reserved.
  *
  * This software is licensed under terms that can be found in the LICENSE file
  * in the root directory of this software component.
  * If no LICENSE file comes with this software, it is provided AS-IS.
  *
  ******************************************************************************
  */
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------*/
#include "main.h"

/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */

/* USER CODE END Includes */

/* Private typedef -----------------------------------------------------------*/
/* USER CODE BEGIN PTD */

/* USER CODE END PTD */

/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */

/* USER CODE END PD */
```

```c
/* Private macro -------------------------------------------------------------*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */

/* Private variables ---------------------------------------------------------*/
TIM_HandleTypeDef htim2;

/* USER CODE BEGIN PV */
uint8_t tim2_count = 0;  // 新增：用于累计定时器中断次数
/* USER CODE END PV */

/* Private function prototypes -----------------------------------------------*/
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_TIM2_Init(void);
/* USER CODE BEGIN PFP */

/* USER CODE END PFP */

/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */

/* USER CODE END 0 */

/**
  * @brief  The application entry point.
  * @retval int
  */
int main(void)
{

  /* USER CODE BEGIN 1 */

  /* USER CODE END 1 */

  /* MCU Configuration--------------------------------------------------------*/

  /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
  HAL_Init();

  /* USER CODE BEGIN Init */

  /* USER CODE END Init */

  /* Configure the system clock */
  SystemClock_Config();

  /* USER CODE BEGIN SysInit */

  /* USER CODE END SysInit */

  /* Initialize all configured peripherals */
```

```
  MX_GPIO_Init();
  MX_TIM2_Init();
  /* USER CODE BEGIN 2 */
  // 新增：启动 TIM2 定时器并使能更新中断
  HAL_TIM_Base_Start_IT(&htim2);
  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
  }
  /* USER CODE END 3 */
}

/**
  * @brief System Clock Configuration
  * @retval None
  */
void SystemClock_Config(void)
{
  RCC_OscInitTypeDef RCC_OscInitStruct = {0};
  RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};

  /** Initializes the RCC Oscillators according to the specified parameters
  * in the RCC_OscInitTypeDef structure.
  */
  RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSE;
  RCC_OscInitStruct.HSEState = RCC_HSE_ON;
  RCC_OscInitStruct.HSEPredivValue = RCC_HSE_PREDIV_DIV1;
  RCC_OscInitStruct.HSIState = RCC_HSI_ON;
  RCC_OscInitStruct.PLL.PLLState = RCC_PLL_ON;
  RCC_OscInitStruct.PLL.PLLSource = RCC_PLLSOURCE_HSE;
  RCC_OscInitStruct.PLL.PLLMUL = RCC_PLL_MUL9;
  if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
  {
    Error_Handler();
  }

  /** Initializes the CPU, AHB and APB buses clocks
  */
  RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                              |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
  RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_PLLCLK;
  RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
  RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV2;
  RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

  if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_2) != HAL_OK)
  {
```

```
    Error_Handler();
  }
}

/**
  * @brief TIM2 Initialization Function
  * @param None
  * @retval None
  */
static void MX_TIM2_Init(void)
{

  /* USER CODE BEGIN TIM2_Init 0 */

  /* USER CODE END TIM2_Init 0 */

  TIM_ClockConfigTypeDef sClockSourceConfig = {0};
  TIM_MasterConfigTypeDef sMasterConfig = {0};

  /* USER CODE BEGIN TIM2_Init 1 */

  /* USER CODE END TIM2_Init 1 */
  htim2.Instance = TIM2;
  htim2.Init.Prescaler = 7199;
  htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
  htim2.Init.Period = 9999; // 修改：原为65535，改为9999才能实现1秒定时
  htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
  htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
  if (HAL_TIM_Base_Init(&htim2) != HAL_OK)
  {
    Error_Handler();
  }
  sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
  if (HAL_TIM_ConfigClockSource(&htim2, &sClockSourceConfig) != HAL_OK)
  {
    Error_Handler();
  }
  sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
  sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
  if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN TIM2_Init 2 */
  // 新增：使能 TIM2 的 NVIC 中断（如果你在CubeMX里没勾选中断，这里必须加上）
  HAL_NVIC_SetPriority(TIM2_IRQn, 0, 0);
  HAL_NVIC_EnableIRQ(TIM2_IRQn);
  /* USER CODE END TIM2_Init 2 */

}

/**
  * @brief GPIO Initialization Function
```

```
  * @param None
  * @retval None
  */
static void MX_GPIO_Init(void)
{
  GPIO_InitTypeDef GPIO_InitStruct = {0};
  /* USER CODE BEGIN MX_GPIO_Init_1 */

  /* USER CODE END MX_GPIO_Init_1 */

  /* GPIO Ports Clock Enable */
  __HAL_RCC_GPIOD_CLK_ENABLE();
  __HAL_RCC_GPIOA_CLK_ENABLE();

  /*Configure GPIO pin Output Level */
  HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);

  /*Configure GPIO pin : PA5 */
  GPIO_InitStruct.Pin = GPIO_PIN_5;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

  /* USER CODE BEGIN MX_GPIO_Init_2 */

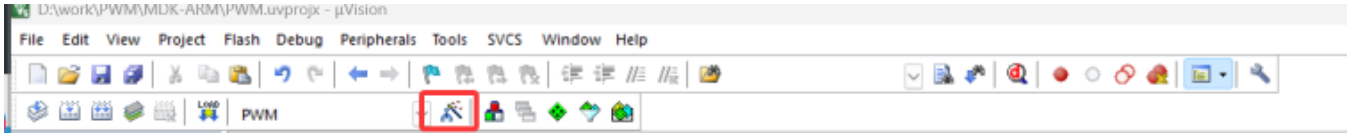
  /* USER CODE END MX_GPIO_Init_2 */
}

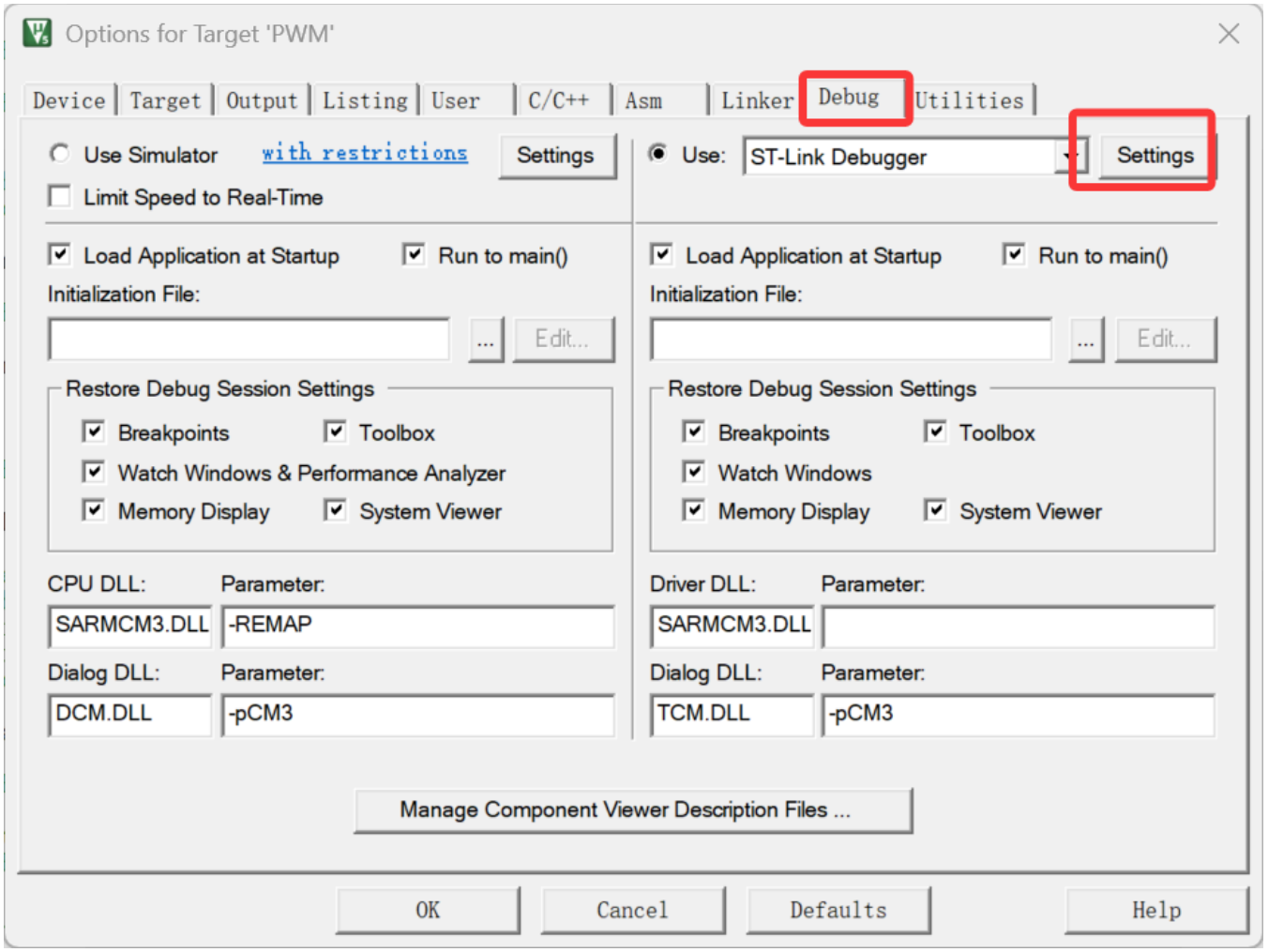
/* USER CODE BEGIN 4 */
/**
  * @brief  定时器更新中断回调函数
  * @param  htim: 定时器句柄
  * @retval None
  */
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
  if (htim->Instance == TIM2)
  {
    tim2_count++;
    if (tim2_count >= 2)  // 累计2次中断，即2秒
    {
      tim2_count = 0;     // 清零计数器
      // 翻转 PA5 引脚电平
      HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    }
  }
}
/* USER CODE END 4 */

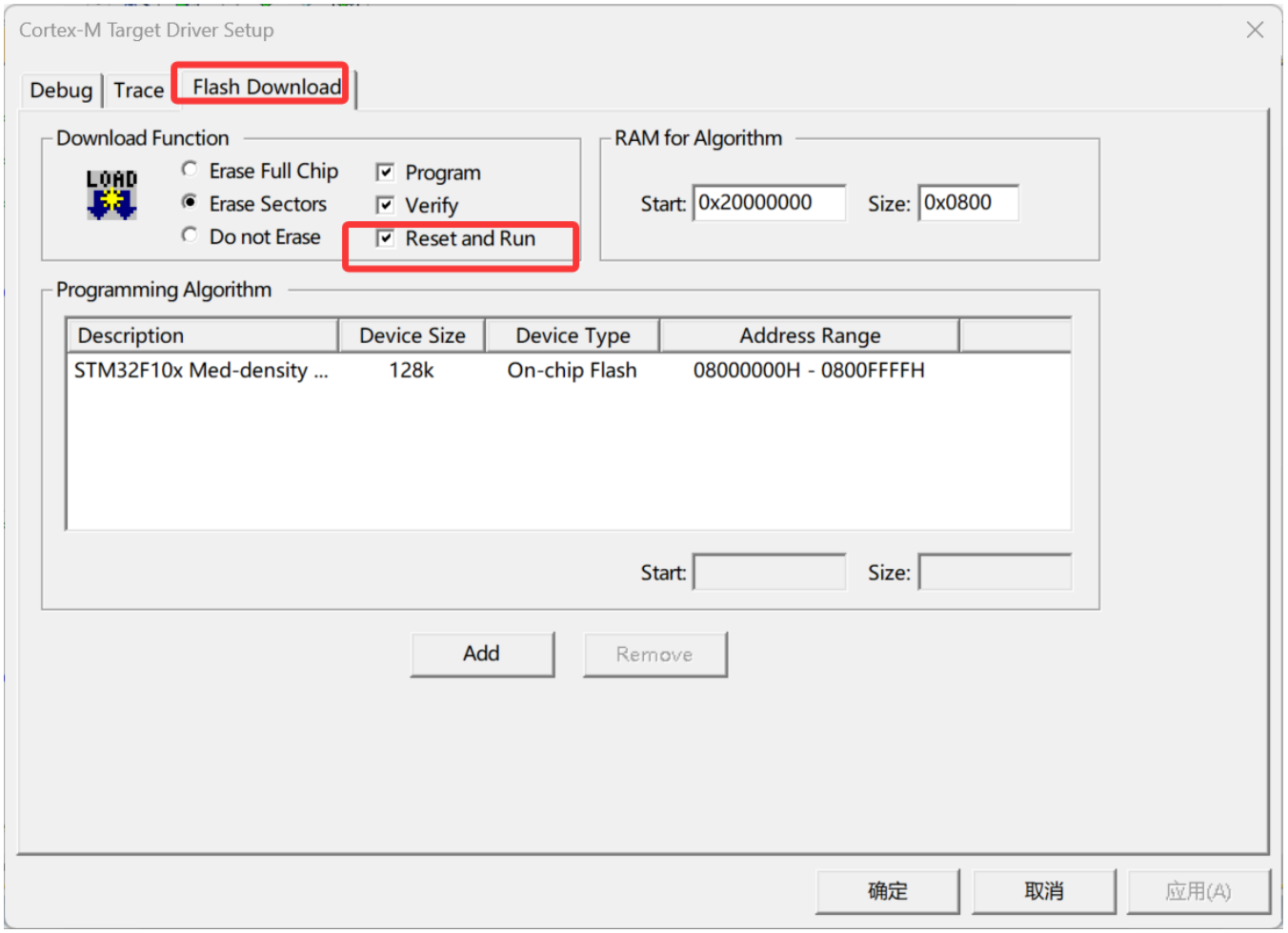
/**
  * @brief  This function is executed in case of error occurrence.
  * @retval None
```

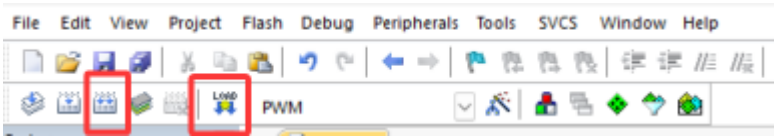
```
  */
void Error_Handler(void)
{
  /* USER CODE BEGIN Error_Handler_Debug */
  /* User can add his own implementation to report the HAL error return state */
  __disable_irq();
  while (1)
  {
  }
  /* USER CODE END Error_Handler_Debug */
}
#ifdef USE_FULL_ASSERT
/**
  * @brief  Reports the name of the source file and the source line number
  *         where the assert_param error has occurred.
  * @param  file: pointer to the source file name
  * @param  line: assert_param error line source number
  * @retval None
  */
void assert_failed(uint8_t *file, uint32_t line)
{
  /* USER CODE BEGIN 6 */
  /* User can add his own implementation to report the file name and line number,
     ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
  /* USER CODE END 6 */
}
#endif /* USE_FULL_ASSERT */
```

3.编译运行




Options for Target 'PWM'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Use Simulator
with restrictions
Settings
Limit Speed to Real-Time
Use:
ST-Link Debugger
Settings
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows & Performance Analyzer
Memory Display
System Viewer
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows
Memory Display
System Viewer
CPU DLL:
Parameter:
SARMCM3.DLL
-REMAP
Dialog DLL:
Parameter:
DCM.DLL
-pCM3
Driver DLL:
Parameter:
SARMCM3.DLL
Dialog DLL:
Parameter:
TCM.DLL
-pCM3
Manage Component Viewer Description Files ...
OK
Cancel
Defaults
Help


Cortex-M Target Driver Setup
Debug
Trace
Flash Download
Download Function
LOAD
Erase Full Chip
Erase Sectors
Do not Erase
Program
Verify
Reset and Run
RAM for Algorithm
Start: 0x20000000
Size: 0x0800
Programming Algorithm
Description
Device Size
Device Type
Address Range
STM32F10x Med-density ...
128k
On-chip Flash
08000000H - 0800FFFFH
Start:
Size:
Add
Remove
确定
取消
应用(A)




```
:ompiling stm32f1xx_hal_rcc.c...
:ompiling stm32f1xx_hal_pwr.c...
:ompiling stm32f1xx_hal_exti.c...
:ompiling stm32f1xx_hal_flash_ex.c...
:ompiling stm32f1xx_hal_flash.c...
:ompiling system_stm32f1xx.c...
.inking...
'rogram Size: Code=4036 RO-data=304 RW-data=20 ZI-data=1708
'romELF: creating hex file...
'TIMELED1\TIMELED1.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:11
```

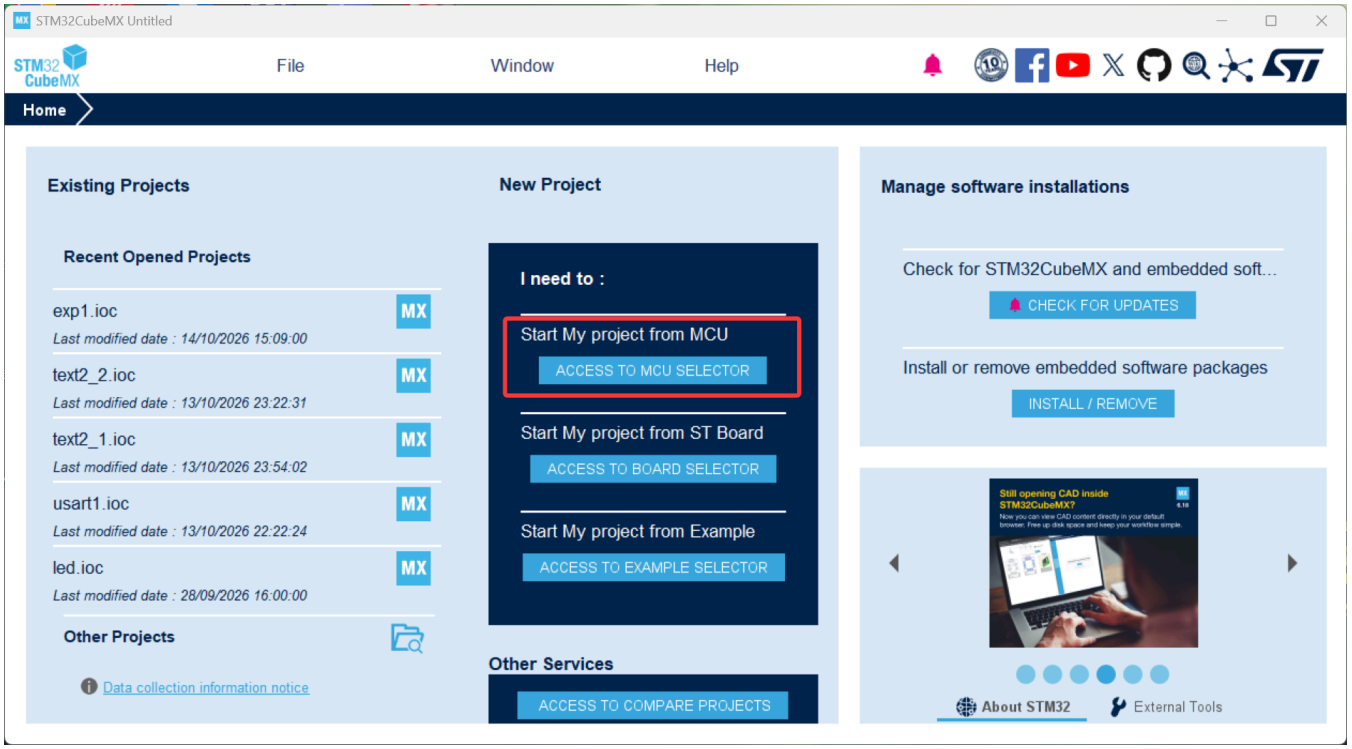
Build Output

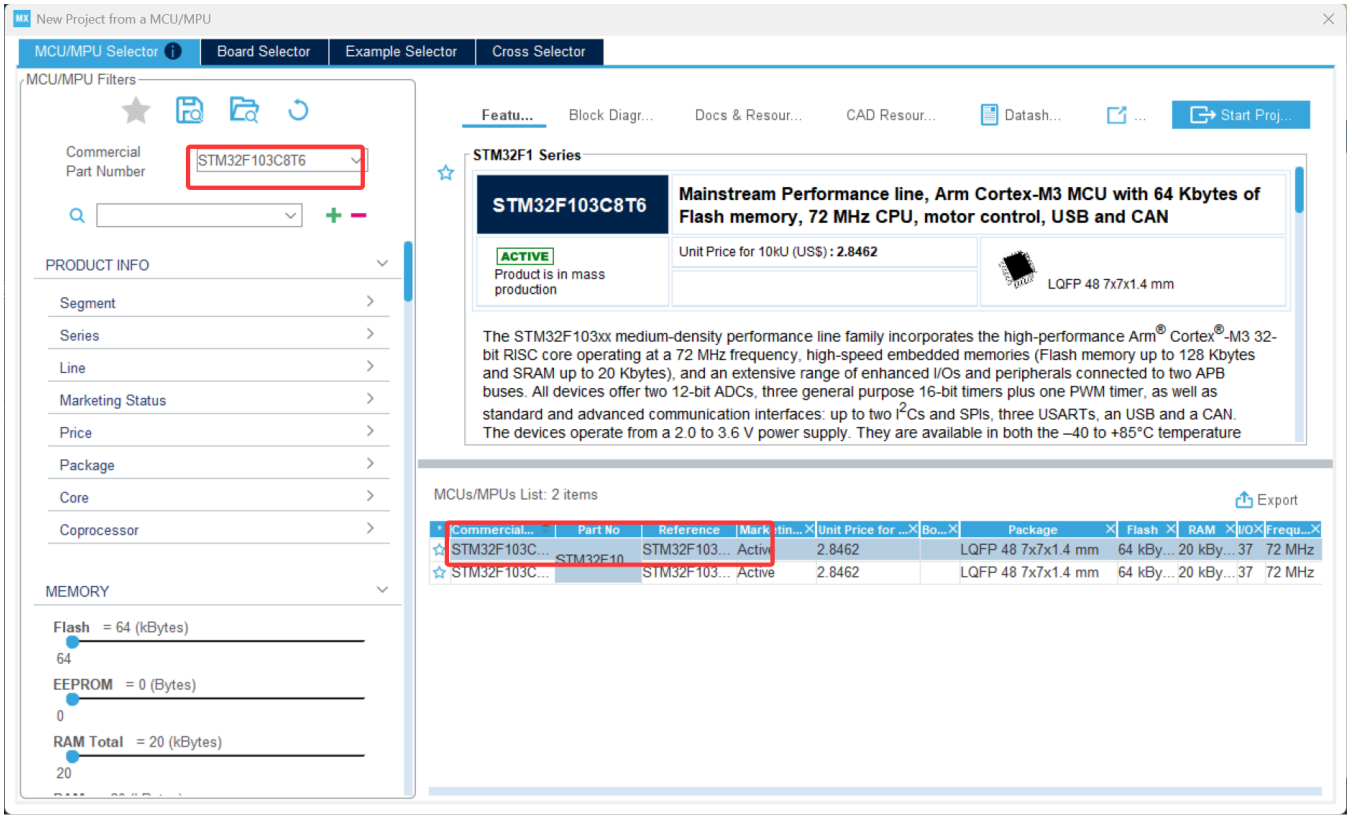
```
compiling stm32f1xx_hal_exti.c...
linking...
Program Size: Code=4036 RO-data=304 RW-data=20 ZI-data=1708
FromELF: creating hex file...
"TIMELED1\TIMELED1.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:11
Load "TIMELED1\\TIMELED1.axf"
Erase Done.
Programming Done.
Verify OK.
Application running ...
Flash Load finished at 21:32:16
```

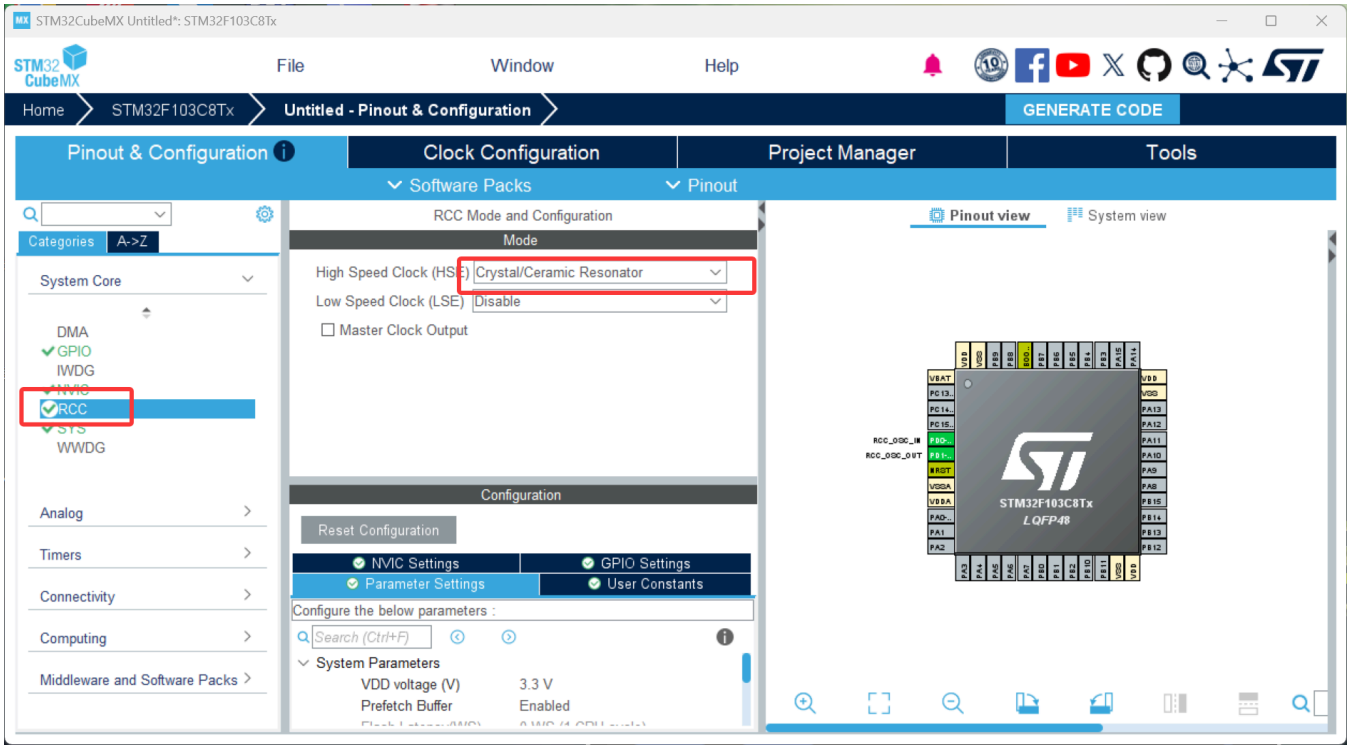
4.实验结果

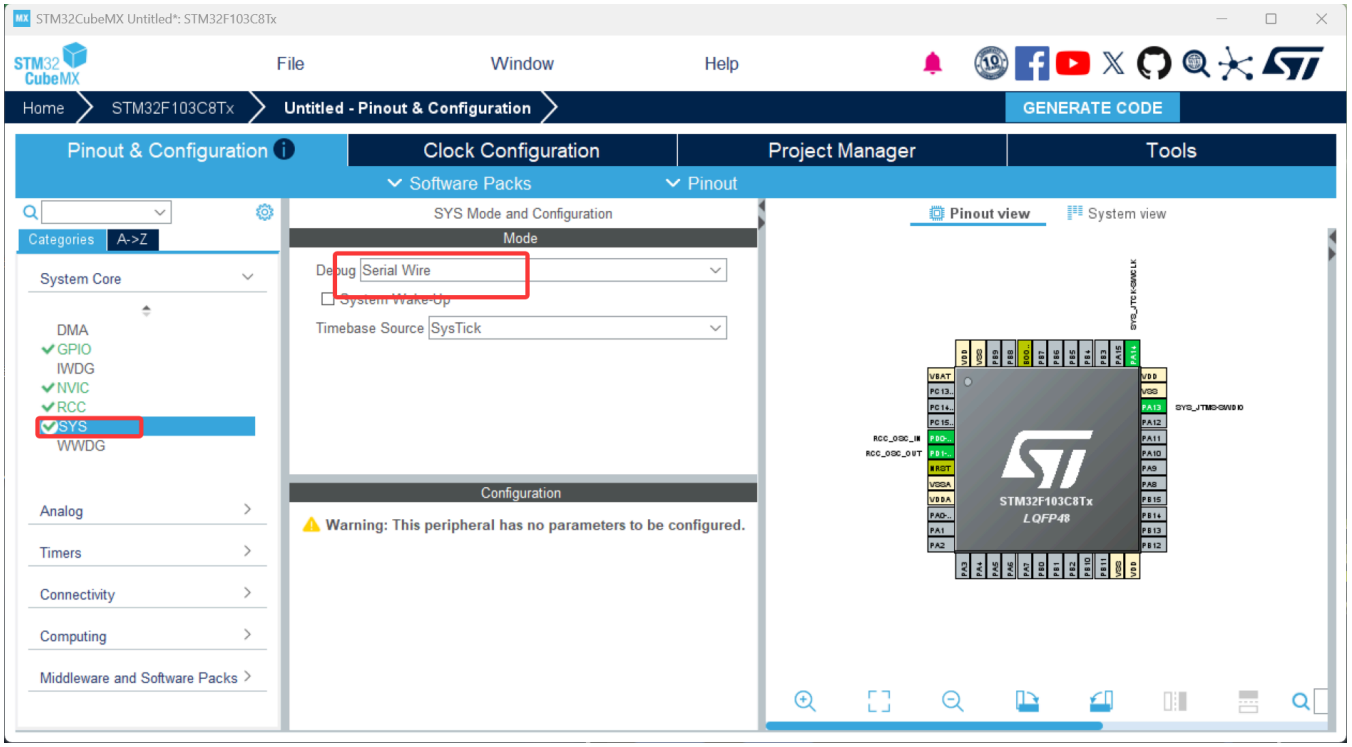
四、PWM模式实现呼吸灯

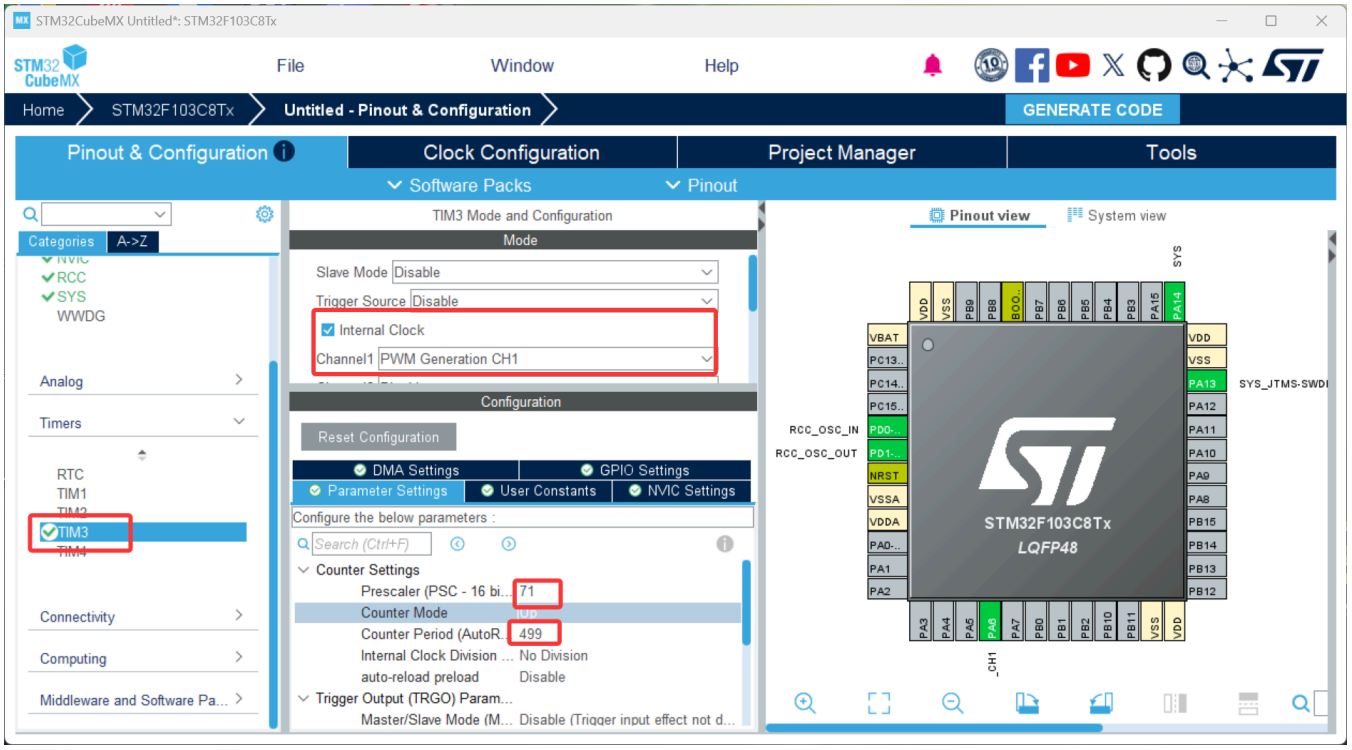
1.创建工程

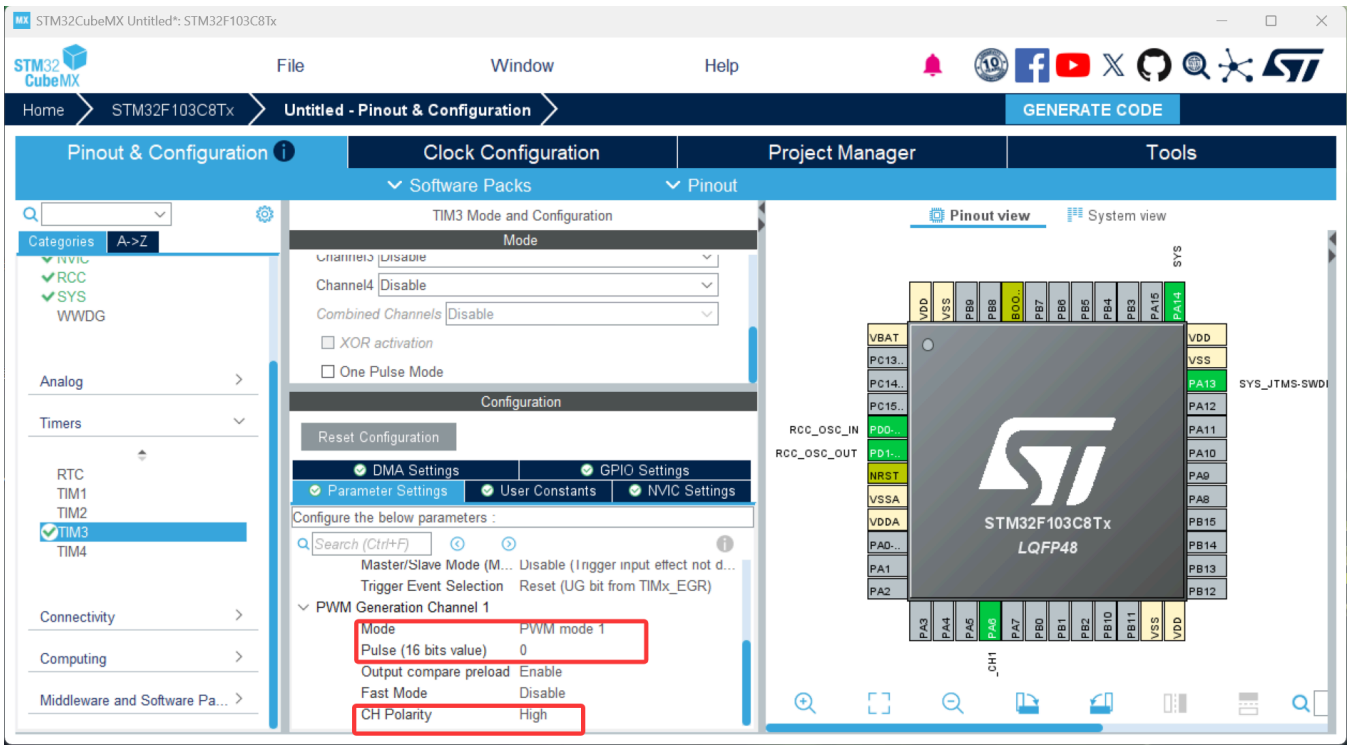


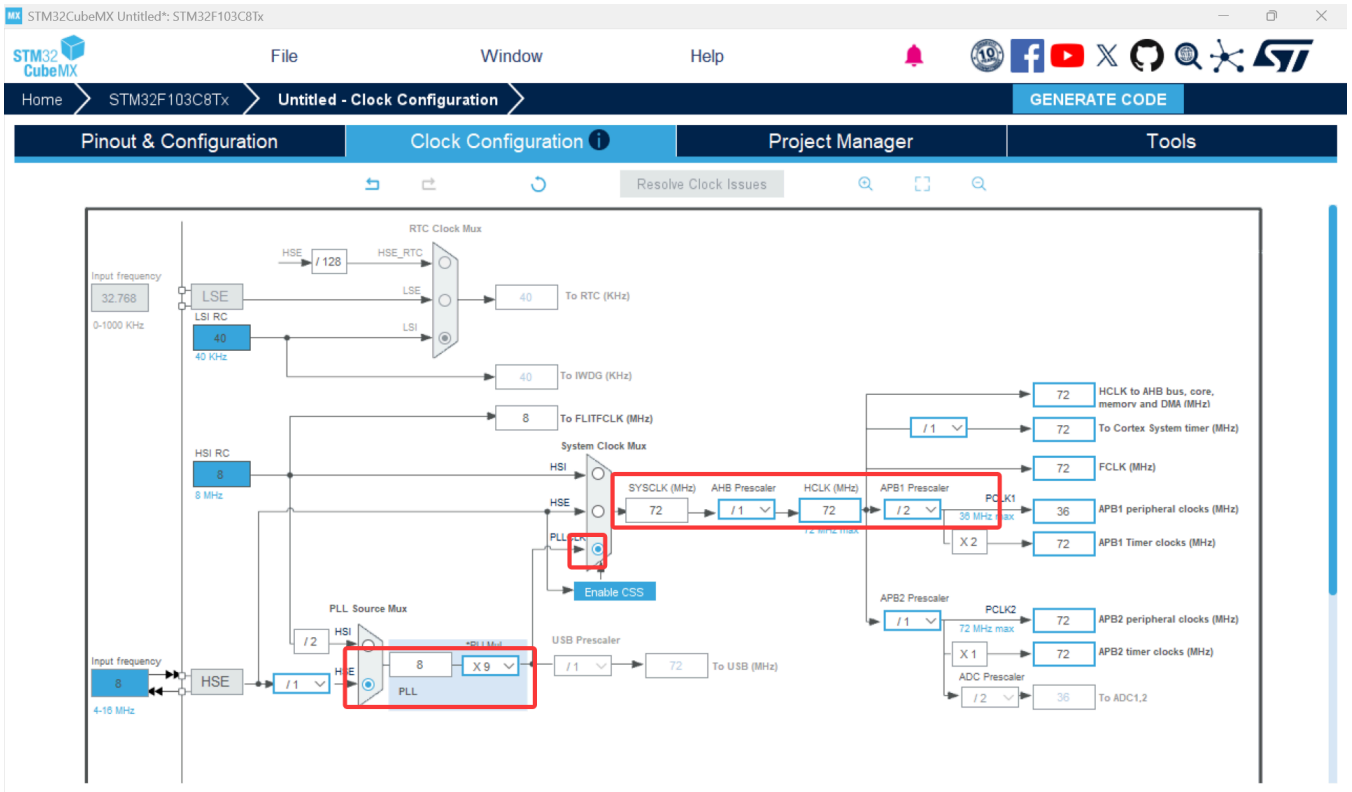

New Project from a MCU/MPU
MCU/MPU Selector
Board Selector
Example Selector
Cross Selector
MCU/MPU Filters
Commercial Part Number
STM32F103C8T6
PRODUCT INFO
Segment
Series
Line
Marketing Status
Price
Package
Core
Coprocessor
MEMORY
Flash = 64 (kBytes)
64
EEPROM = 0 (Bytes)
0
RAM Total = 20 (kBytes)
20
Featu...
Block Diagr...
Docs & Resour...
CAD Resour...
Datash...
Start Proj...
STM32F1 Series
STM32F103C8T6
Mainstream Performance line, Arm Cortex-M3 MCU with 64 Kbytes of Flash memory, 72 MHz CPU, motor control, USB and CAN
ACTIVE
Product is in mass production
Unit Price for 10kU (US$) : 2.8462
LQFP 48 7x7x1.4 mm
The STM32F103xx medium-density performance line family incorporates the high-performance Arm® Cortex®-M3 32-bit RISC core operating at a 72 MHz frequency, high-speed embedded memories (Flash memory up to 128 Kbytes and SRAM up to 20 Kbytes), and an extensive range of enhanced I/Os and peripherals connected to two APB buses. All devices offer two 12-bit ADCs, three general purpose 16-bit timers plus one PWM timer, as well as standard and advanced communication interfaces: up to two I2Cs and SPIs, three USARTs, an USB and a CAN. The devices operate from a 2.0 to 3.6 V power supply. They are available in both the –40 to +85°C temperature
MCUs/MPUs List: 2 items
Export
Commercial...
Part No
Reference
Unit Price for ...
Package
Flash
RAM
I/O
Frequ...
STM32F103C...
STM32F10
STM32F103...
Active
2.8462
LQFP 48 7x7x1.4 mm
64 kBy...
20 kBy...
37
72 MHz

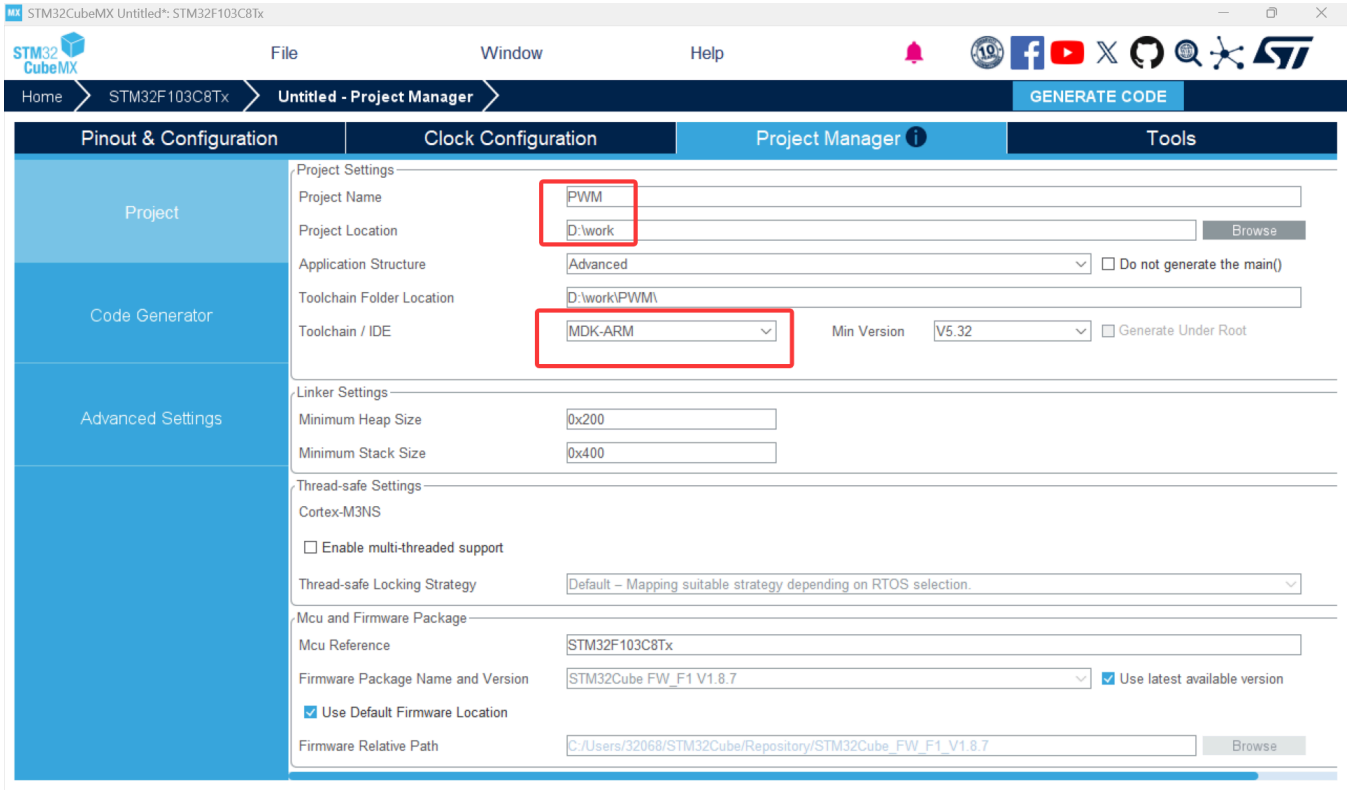

STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
Home
STM32F103C8Tx
Untitled - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Categories
A->Z
System Core
DMA
GPIO
IWDG
RCC
WWDG
Analog
Timers
Connectivity
Computing
Middleware and Software Packs
RCC Mode and Configuration
Mode
High Speed Clock (HSE)
Crystal/Ceramic Resonator
Low Speed Clock (LSE)
Disable
Master Clock Output
Configuration
Reset Configuration
NVIC Settings
GPIO Settings
Parameter Settings
User Constants
Configure the below parameters :
Search (Ctrl+F)
System Parameters
VDD voltage (V)
3.3 V
Prefetch Buffer
Enabled
Pinout view
System view
STM32F103C8Tx
LQFP48

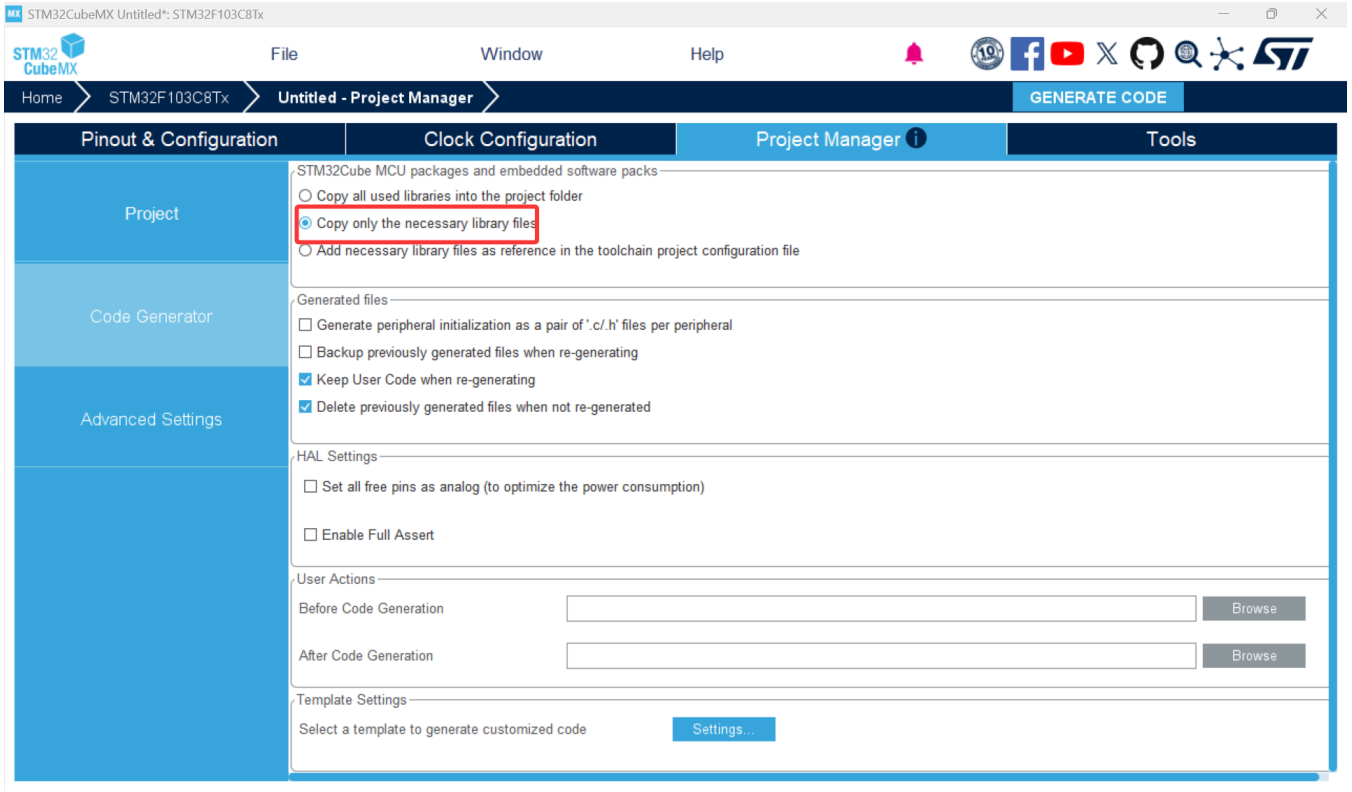

STM32CubeMX Untitled*: STM32F103C8Tx
File
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Help
Home
STM32F103C8Tx
Untitled - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Categories
A->Z
System Core
DMA
GPIO
IWDG
NVIC
RCC
SYS
WWDG
Analog
Timers
Connectivity
Computing
Middleware and Software Packs
SYS Mode and Configuration
Mode
Debug Serial Wire
System Wake-Up
Timebase Source SysTick
Configuration
Warning: This peripheral has no parameters to be configured.
Pinout view
System view
STM32F103C8Tx
LQFP48


STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
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STM32F103C8Tx
Untitled - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Categories
A->Z
RCC
SYS
WWDG
Analog
Timers
RTC
TIM1
TIM3
Connectivity
Computing
TIM3 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Internal Clock
Channel1 PWM Generation CH1
Configuration
Reset Configuration
DMA Settings
GPIO Settings
Parameter Settings
User Constants
NVIC Settings
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bi... 71
Counter Mode
Counter Period (AutoR... 499
Internal Clock Division ... No Division
auto-reload preload Disable
Trigger Output (TRGO) Param...
Master/Slave Mode (M... Disable (Trigger input effect not d...
Pinout view
System view
RCC_OSC_IN
RCC_OSC_OUT
SYS_JTMS-SWDI
STM32F103C8Tx
LQFP48


STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
Home
STM32F103C8Tx
Untitled - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Categories
A->Z
RCC
SYS
WWDG
Analog
Timers
RTC
TIM1
TIM2
TIM3
TIM4
Connectivity
Computing
Middleware and Software Pa...
TIM3 Mode and Configuration
Mode
Channel4 Disable
Combined Channels Disable
XOR activation
One Pulse Mode
Configuration
Reset Configuration
DMA Settings
GPIO Settings
Parameter Settings
User Constants
NVIC Settings
Configure the below parameters :
Search (Ctrl+F)
Master/Slave Mode (M... Disable (Trigger input effect not d...
Trigger Event Selection Reset (UG bit from TIMx_EGR)
PWM Generation Channel 1
Mode PWM mode 1
Pulse (16 bits value) 0
Output compare preload Enable
Fast Mode Disable
CH Polarity High
Pinout view
System view
STM32F103C8Tx
LQFP48
RCC_OSC_IN
RCC_OSC_OUT
SYS_JTMS-SWDI


STM32CubeMX Untitled*: STM32F103C8Tx
File
Window
Help
Home
STM32F103C8Tx
Untitled - Clock Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Resolve Clock Issues
RTC Clock Mux
Input frequency
32.768
0-1000 KHz
LSE
LSI RC
40
40 KHz
HSE
/ 128
HSE_RTC
LSE
LSI
To RTC (KHz)
To IWDG (KHz)
To FLITFCLK (MHz)
HSI RC
8 MHz
System Clock Mux
HSI
HSE
PLLCLK
SYSCLK (MHz)
AHB Prescaler
HCLK (MHz)
APB1 Prescaler
PCLK1
36 MHz max
Enable CSS
PLL Source Mux
USB Prescaler
To USB (MHz)
PLL
X 9
Input frequency
4-16 MHz
APB2 Prescaler
PCLK2
72 MHz max
ADC Prescaler
To ADC1,2
HCLK to AHB bus, core, memory and DMA (MHz)
To Cortex System timer (MHz)
FCLK (MHz)
APB1 peripheral clocks (MHz)
APB1 Timer clocks (MHz)
APB2 peripheral clocks (MHz)
APB2 timer clocks (MHz)





2.代码撰写

(1) 在 `/* USER CODE BEGIN PV */` 之间添加:

```
uint16_t pwm_val = 0;     // 当前占空比值 (0~500)
uint8_t direction = 1;    // 1=渐亮，0=渐暗
```

(2) 在 `/* USER CODE BEGIN 2 */` 之间添加:

```
HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_1);  // 启动TIM3通道1 PWM输出
```

(3) 在 `/* USER CODE BEGIN WHILE */` 和 `/* USER CODE END WHILE */` 之间添加:

```
    // 呼吸速度调节：觉得快就把 5 改成 10 或 15
    HAL_Delay(5);

    if (direction)
    {
      pwm_val++;
      if (pwm_val >= 500) direction = 0; // 到达最亮转暗
    }
    else
    {
      pwm_val--;
      if (pwm_val == 0) direction = 1;   // 到达最暗转亮
    }

    // 动态修改比较寄存器（CCR1）的值，改变PWM占空比
    __HAL_TIM_SetCompare(&htim3, TIM_CHANNEL_1, pwm_val);
```

(4) 完整代码

```
/* USER CODE BEGIN Header */
/**
  ******************************************************************************
  * @file           : main.c
  * @brief          : Main program body
  ******************************************************************************
  * @attention
  *
  * Copyright (c) 2026 STMicroelectronics.
  * All rights reserved.
  *
  * This software is licensed under terms that can be found in the LICENSE file
  * in the root directory of this software component.
  * If no LICENSE file comes with this software, it is provided AS-IS.
  *
  ******************************************************************************
  */
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------*/
#include "main.h"

/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */

/* USER CODE END Includes */

/* Private typedef -----------------------------------------------------------*/
/* USER CODE BEGIN PTD */
```

```c
/* USER CODE END PTD */

/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */

/* USER CODE END PD */

/* Private macro -------------------------------------------------------------*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */

/* Private variables ---------------------------------------------------------*/
TIM_HandleTypeDef htim3;

/* USER CODE BEGIN PV */
uint16_t pwm_val = 0;     // 锟斤拷锟斤拷前占锟秸憋拷锟斤拷值 (锟斤拷锟斤拷锟斤拷围 0 ~ 500)
uint8_t direction = 1;    // 锟斤拷锟斤拷化锟斤拷锟斤拷锟斤拷1=锟斤拷锟斤拷锟斤拷锟斤拷(锟斤拷锟斤拷锟斤拷锟斤拷锟斤拷)锟斤拷锟斤拷0=锟斤拷锟斤拷锟斤拷锟斤拷锟斤拷(锟斤拷锟捷硷拷
锟斤拷)
/* USER CODE END PV */

/* Private function prototypes -----------------------------------------------*/
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_TIM3_Init(void);
/* USER CODE BEGIN PFP */

/* USER CODE END PFP */

/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */

/* USER CODE END 0 */

/**
  * @brief  The application entry point.
  * @retval int
  */
int main(void)
{

  /* USER CODE BEGIN 1 */

  /* USER CODE END 1 */

  /* MCU Configuration--------------------------------------------------------*/

  /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
  HAL_Init();

  /* USER CODE BEGIN Init */
```

```c
  /* USER CODE END Init */

  /* Configure the system clock */
  SystemClock_Config();

  /* USER CODE BEGIN SysInit */

  /* USER CODE END SysInit */

  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_TIM3_Init();
  /* USER CODE BEGIN 2 */
  // 锟斤拷锟斤拷 TIM3 通锟斤拷1 锟斤拷 PWM 锟斤拷锟斤拷锟斤拷锟斤拷锟斤拷锟接?PA6 锟斤拷锟斤拷脚 o 拷
  HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_1);
  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    // 锟斤拷锟斤拷锟斤拷锟斤拷锟劫度碉拷锟节关硷拷锟姐】
    // 锟斤拷锟斤拷锟斤拷锟斤拷煤锟斤拷锟教□拷软□拷锟?5 锟侥筹拷 10 锟斤拷 15锟斤拷锟斤拷锟斤拷锟斤拷越锟斤拷锟斤拷锟斤拷锟皆斤拷锟?
    // 之前锟斤拷 1 锟斤拷 2 锟结导锟斤拷锟今急达拷锟斤拷烁锟斤拷5 锟饺较接斤拷锟剿碉拷锟斤拷锟斤拷锟斤拷锟斤拷锟斤拷锟?
    HAL_Delay(5);

    if (direction)
    {
      pwm_val++; // 占锟秸憋拷锟斤拷锟接 o 拷LED 锟金渐憋拷锟斤拷
      if (pwm_val >= 500)
      {
        direction = 0; // 锟斤拷锟斤拷锟斤拷锟斤拷锟斤拷锟斤拷锟叫伙拷为锟斤拷锟斤拷锟斤拷
      }
    }
    else
    {
      pwm_val--; // 占锟秸比硷拷锟劫 o 拷LED 锟金渐变暗
      if (pwm_val == 0)
      {
        direction = 1; // 锟斤拷锟斤拷锟斤拷锟筋暗锟斤拷锟斤拷叫伙拷为锟斤拷锟斤拷锟斤拷
      }
    }

    // 锟斤拷锟斤拷态锟睫改讹拷时锟斤拷锟斤拷饺较寄达拷锟斤拷锟斤拷锟斤拷CCR1锟斤拷锟斤拷锟斤拷值锟斤拷锟斤拷接讹拷锟侥憋拷 PWM 占锟秸憋拷
    __HAL_TIM_SetCompare(&htim3, TIM_CHANNEL_1, pwm_val);

    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
  }
```

```
  /* USER CODE END 3 */
}

/**
  * @brief System Clock Configuration
  * @retval None
  */
void SystemClock_Config(void)
{
  RCC_OscInitTypeDef RCC_OscInitStruct = {0};
  RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};

  /** Initializes the RCC Oscillators according to the specified parameters
  * in the RCC_OscInitTypeDef structure.
  */
  RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSE;
  RCC_OscInitStruct.HSEState = RCC_HSE_ON;
  RCC_OscInitStruct.HSEPredivValue = RCC_HSE_PREDIV_DIV1;
  RCC_OscInitStruct.HSIState = RCC_HSI_ON;
  RCC_OscInitStruct.PLL.PLLState = RCC_PLL_ON;
  RCC_OscInitStruct.PLL.PLLSource = RCC_PLLSOURCE_HSE;
  RCC_OscInitStruct.PLL.PLLMUL = RCC_PLL_MUL9;
  if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
  {
    Error_Handler();
  }

  /** Initializes the CPU, AHB and APB buses clocks
  */
  RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                              |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
  RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_PLLCLK;
  RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
  RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV2;
  RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

  if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_2) != HAL_OK)
  {
    Error_Handler();
  }
}

/**
  * @brief TIM3 Initialization Function
  * @param None
  * @retval None
  */
static void MX_TIM3_Init(void)
{

  /* USER CODE BEGIN TIM3_Init 0 */

  /* USER CODE END TIM3_Init 0 */
```

```
  TIM_ClockConfigTypeDef sClockSourceConfig = {0};
  TIM_MasterConfigTypeDef sMasterConfig = {0};
  TIM_OC_InitTypeDef sConfigOC = {0};

  /* USER CODE BEGIN TIM3_Init 1 */

  /* USER CODE END TIM3_Init 1 */
  htim3.Instance = TIM3;
  htim3.Init.Prescaler = 71;
  htim3.Init.CounterMode = TIM_COUNTERMODE_UP;
  htim3.Init.Period = 65535;
  htim3.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
  htim3.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
  if (HAL_TIM_Base_Init(&htim3) != HAL_OK)
  {
    Error_Handler();
  }
  sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
  if (HAL_TIM_ConfigClockSource(&htim3, &sClockSourceConfig) != HAL_OK)
  {
    Error_Handler();
  }
  if (HAL_TIM_PWM_Init(&htim3) != HAL_OK)
  {
    Error_Handler();
  }
  sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
  sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
  if (HAL_TIMEx_MasterConfigSynchronization(&htim3, &sMasterConfig) != HAL_OK)
  {
    Error_Handler();
  }
  sConfigOC.OCMode = TIM_OCMODE_PWM1;
  sConfigOC.Pulse = 0;
  sConfigOC.OCPolarity = TIM_OCPOLARITY_HIGH;
  sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;
  if (HAL_TIM_PWM_ConfigChannel(&htim3, &sConfigOC, TIM_CHANNEL_1) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN TIM3_Init 2 */

  /* USER CODE END TIM3_Init 2 */
  HAL_TIM_MspPostInit(&htim3);

}

/**
  * @brief GPIO Initialization Function
  * @param None
  * @retval None
  */
```

```c
static void MX_GPIO_Init(void)
{
  /* USER CODE BEGIN MX_GPIO_Init_1 */

  /* USER CODE END MX_GPIO_Init_1 */

  /* GPIO Ports Clock Enable */
  __HAL_RCC_GPIOD_CLK_ENABLE();
  __HAL_RCC_GPIOA_CLK_ENABLE();

  /* USER CODE BEGIN MX_GPIO_Init_2 */

  /* USER CODE END MX_GPIO_Init_2 */
}

/* USER CODE BEGIN 4 */

/* USER CODE END 4 */

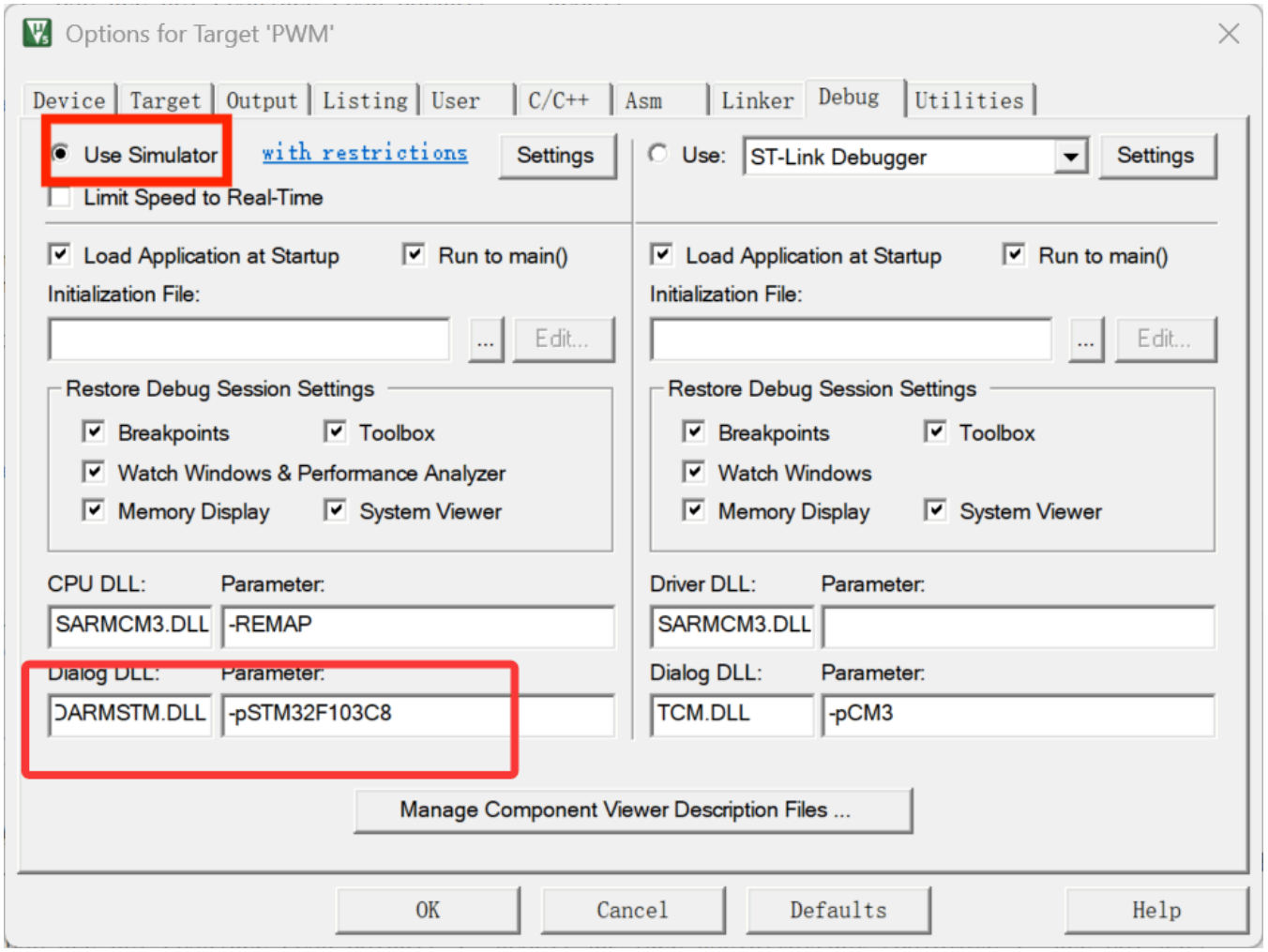
/**
  * @brief  This function is executed in case of error occurrence.
  * @retval None
  */
void Error_Handler(void)
{
  /* USER CODE BEGIN Error_Handler_Debug */
  /* User can add his own implementation to report the HAL error return state */
  __disable_irq();
  while (1)
  {
  }
  /* USER CODE END Error_Handler_Debug */
}
#ifdef USE_FULL_ASSERT
/**
  * @brief  Reports the name of the source file and the source line number
  *         where the assert_param error has occurred.
  * @param  file: pointer to the source file name
  * @param  line: assert_param error line source number
  * @retval None
  */
void assert_failed(uint8_t *file, uint32_t line)
{
  /* USER CODE BEGIN 6 */
  /* User can add his own implementation to report the file name and line number,
     ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
  /* USER CODE END 6 */
}
#endif /* USE_FULL_ASSERT */
```

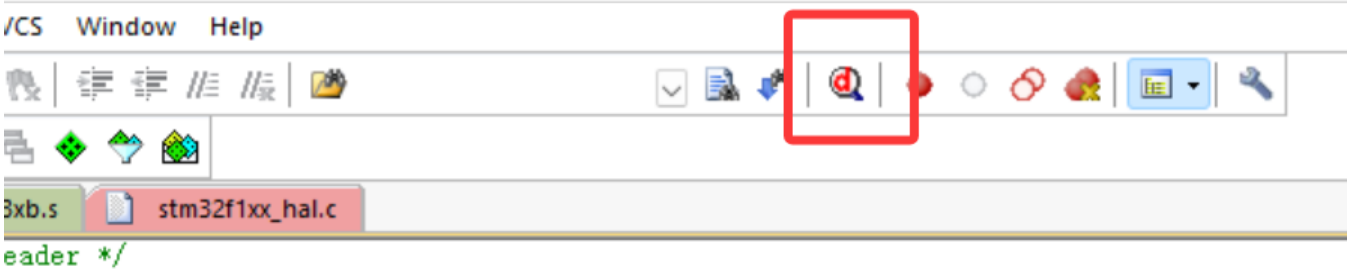
3.代码编译 运行如上

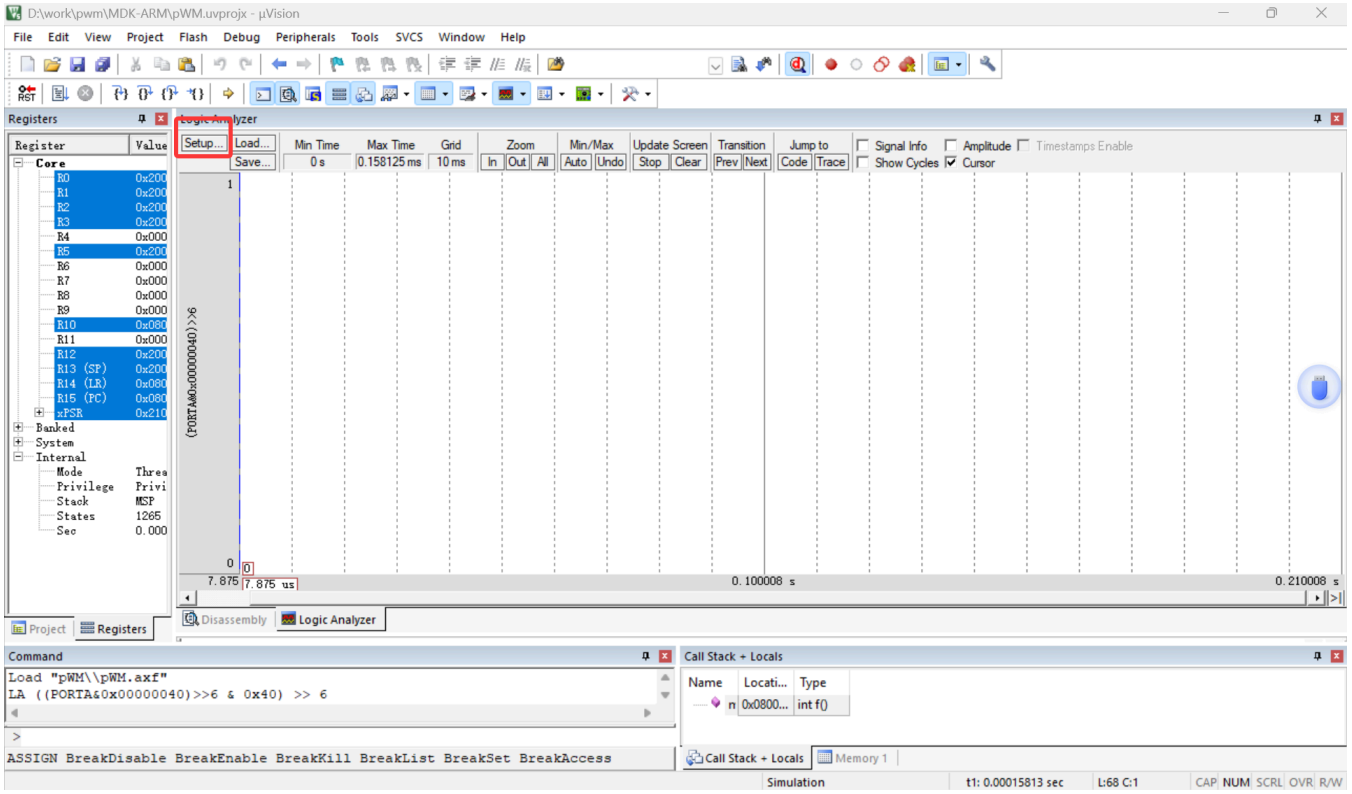
4.实验结果

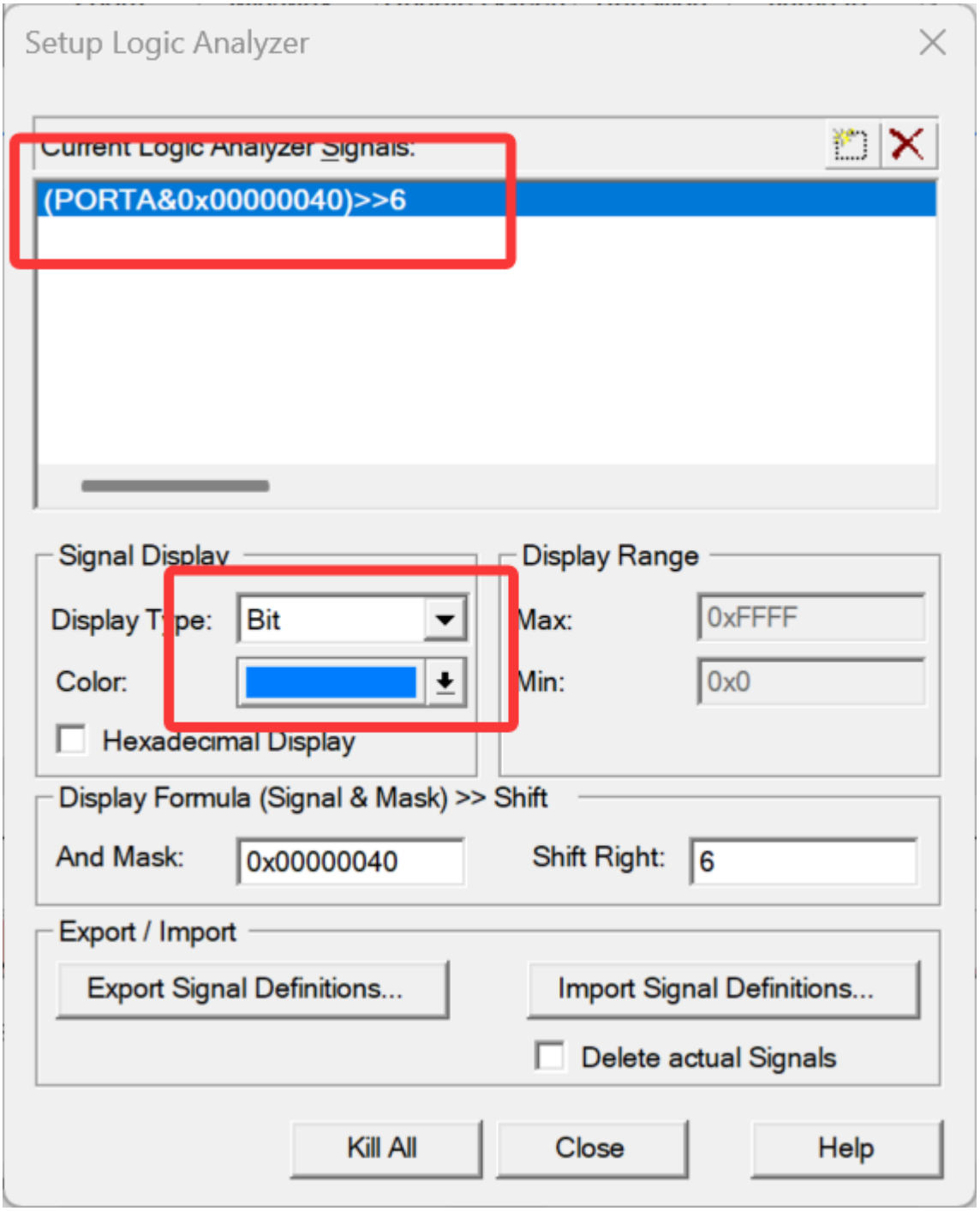
5.PWM输出波形

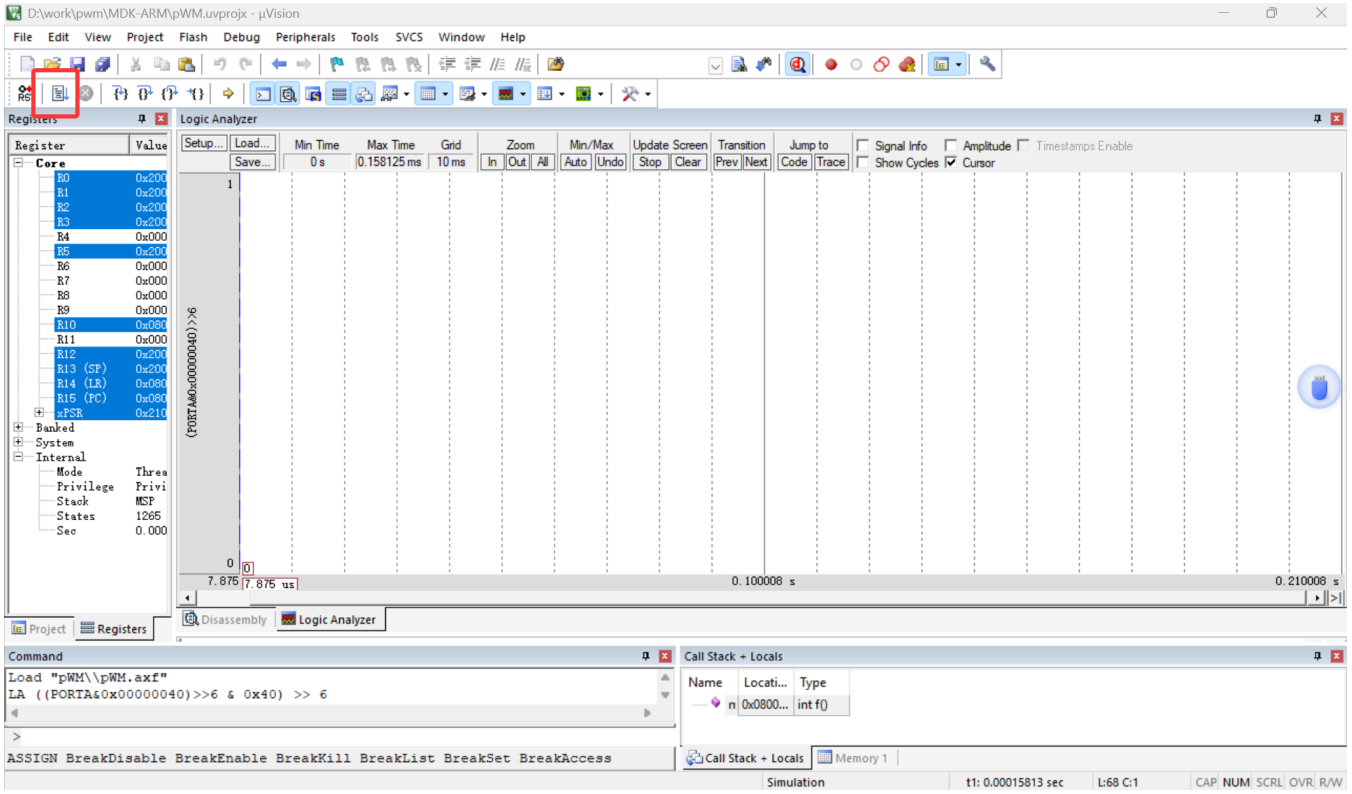
（1）环境搭建


Options for Target 'PWM'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Use Simulator
with restrictions
Settings
Limit Speed to Real-Time
Use:
ST-Link Debugger
Settings
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows & Performance Analyzer
Watch Windows
Memory Display
System Viewer
CPU DLL:
Parameter:
SARMCM3.DLL
-REMAP
Driver DLL:
Dialog DLL:
DARMSTM.DLL
-pSTM32F103C8
TCM.DLL
-pCM3
Manage Component Viewer Description Files ...
OK
Cancel
Defaults
Help

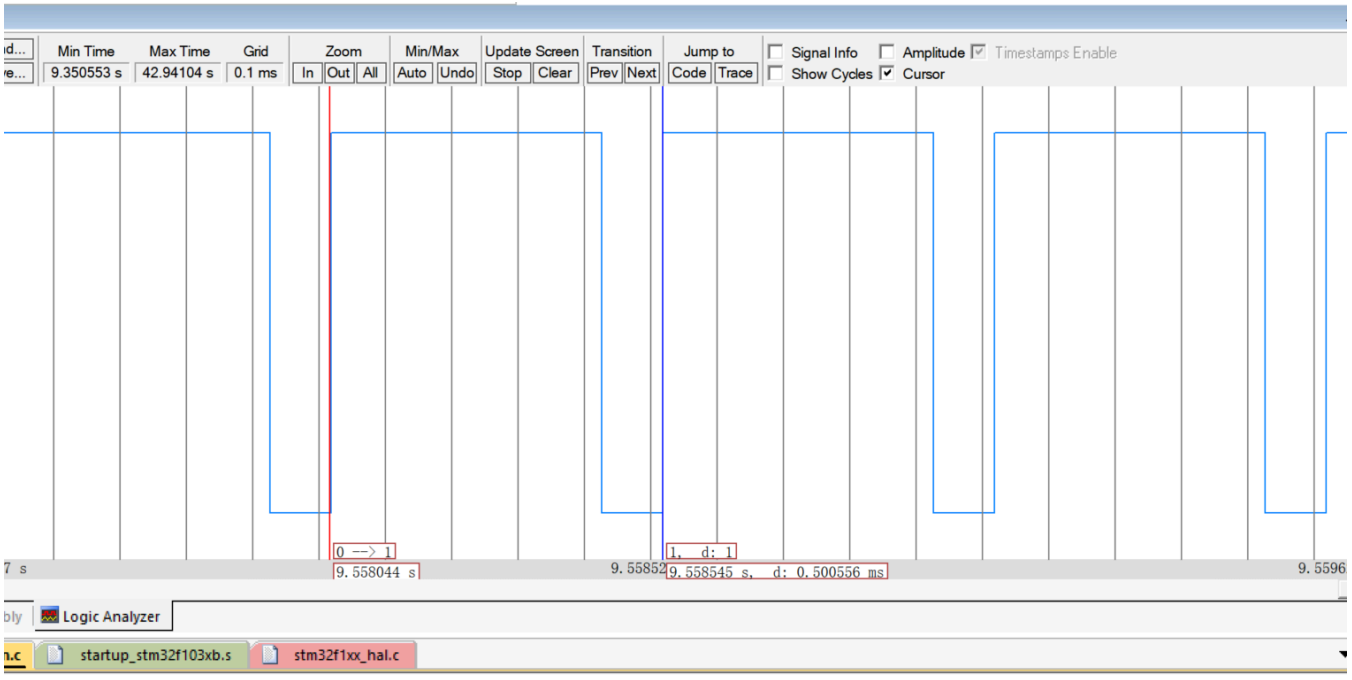

VCS
Window
Help
3xb.s
stm32f1xx_hal.c
eader */

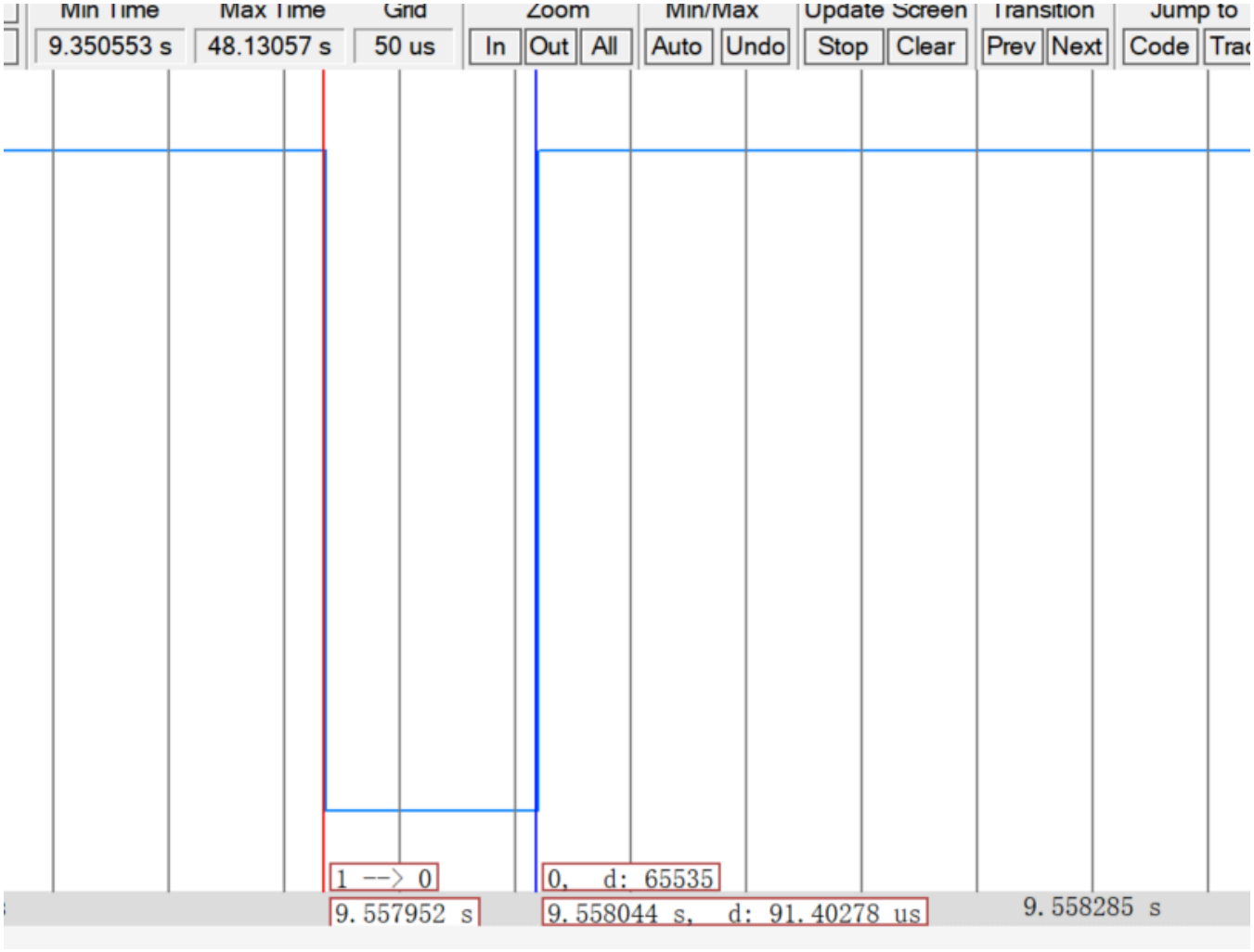

D:\work\pwm\MDK-ARM\pWM.uvprojx - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x200
R1 0x200
R2 0x200
R3 0x200
R4 0x000
R5 0x200
R6 0x000
R7 0x000
R8 0x000
R9 0x000
R10 0x080
R11 0x000
R12 0x200
R13 (SP) 0x200
R14 (LR) 0x080
R15 (PC) 0x080
xPSR 0x210
Banked
System
Internal
Mode Threa
Privilege Privi
Stack MSP
States 1265
Sec 0.000
Logic Analyzer
Setup... Load... Save...
Min Time 0 s
Max Time 0.158125 ms
Grid 10 ms
Zoom In Out All
Min/Max Auto Undo
Update Screen Stop Clear
Transition Prev Next
Jump to Code Trace
Signal Info
Show Cycles
Amplitude
Cursor
Timestamps Enable
(PORTA&0x00000040)>>6
1
0
7.875
7.875 us
0.100008 s
0.210008 s
Project Registers
Disassembly Logic Analyzer
Command
Load "pWM\\pWM.axf"
LA ((PORTA&0x00000040)>>6 & 0x40) >> 6
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess
Call Stack + Locals
Name Locati... Type
n 0x0800... int f()
Call Stack + Locals Memory 1
Simulation t1: 0.00015813 sec L:68 C:1 CAP NUM SCRL OVR R/W


Setup Logic Analyzer
Current Logic Analyzer Signals:
(PORTA&0x00000040)>>6
Signal Display
Display Type:
Bit
Color:
Hexadecimal Display
Display Range
Max:
0xFFFF
Min:
0x0
Display Formula (Signal & Mask) >> Shift
And Mask:
0x00000040
Shift Right:
6
Export / Import
Export Signal Definitions...
Import Signal Definitions...
Delete actual Signals
Kill All
Close
Help


D:\work\pwm\MDK-ARM\pWM.uvprojx - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x200
R1 0x200
R2 0x200
R3 0x200
R4 0x000
R5 0x200
R6 0x000
R7 0x000
R8 0x000
R9 0x000
R10 0x080
R11 0x000
R12 0x200
R13 (SP) 0x200
R14 (LR) 0x080
R15 (PC) 0x080
xPSR 0x210
Banked
System
Internal
Mode Threa
Privilege Privi
Stack MSP
States 1265
Sec 0.000
Logic Analyzer
Setup... Load... Save...
Min Time 0 s
Max Time 0.158125 ms
Grid 10 ms
Zoom In Out All
Min/Max Auto Undo
Update Screen Stop Clear
Transition Prev Next
Jump to Code Trace
Signal Info
Show Cycles
Amplitude
Cursor
Timestamps Enable
(PORTA&0x00000040)>>6
1
0
7.875 7.875 us
0.100008 s
0.210008 s
Project Registers
Disassembly Logic Analyzer
Command
Load "pWM\\pWM.axf"
LA ((PORTA&0x00000040)>>6 & 0x40) >> 6
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess
Call Stack + Locals
Name Locati... Type
n 0x0800... int f()
Call Stack + Locals Memory 1
Simulation t1: 0.00015813 sec L:68 C:1 CAP NUM SCRL OVR R/W

（2）分析波形图


Min Time 9.350553 s
Max Time 42.94104 s
Grid 0.1 ms
Zoom In Out All
Min/Max Auto Undo
Update Screen Stop Clear
Transition Prev Next
Jump to Code Trace
Signal Info
Show Cycles
Amplitude
Cursor
Timestamps Enable
0 --> 1
1, d: 1
9.558044 s
9.558545 s, d: 0.500556 ms
Logic Analyzer
startup_stm32f103xb.s
stm32f1xx_hal.c



PWM 周期整体观测

下图为呼吸灯运行过程中 PWM 波形的整体观测，通过光标测量得到单个 PWM 周期时长为 **0.500556 ms**。

周期参数验证：

- 理论计算值：$T = \frac{(ARR+1)\times(PSC+1)}{f_{clk}} = \frac{(499+1)\times(71+1)}{72\times10^6} = 0.5\ \text{ms}$
- 实际仿真值：约 0.5006 ms

实测值与理论值误差小于 0.2%，验证了定时器预分频、自动重装载参数配置正确，PWM 输出频率稳定为 2kHz，高于人眼视觉暂留频率，保证 LED 亮度调节无频闪。

脉宽细节与占空比观测

图为波形放大后的脉宽测量，测得该时刻低电平持续时间为 **91.40278 μs**。

占空比计算验证：已知周期 $T \approx 500.556\ \mu\text{s}$，低电平时间 $T_L \approx 91.403\ \mu\text{s}$，则：高电平时间：

$T_H = T - T_L \approx 500.556 - 91.403 = 409.153\ \mu\text{s}$ 占空比：

$Duty = \frac{T_H}{T} \times 100\% \approx \frac{409.153}{500.556} \times 100\% \approx 81.7\%$

该占空比对应呼吸灯 “渐亮 / 渐暗” 过程中的某一中间状态，验证了程序可以通过 `__HAL_TIM_SetCompare()` 动态修改比较寄存器值，连续调整 PWM 占空比，进而实现 LED 亮度的平滑过渡。

呼吸过程整体验证

连续观测波形可看到：PWM 的高电平宽度随时间线性递增，达到最大值后线性递减，周期性循环变化。完整的“0%→100%→0%” 占空比变化周期约为 5 秒，与程序中 “5ms 步进 × 500 级亮度 × 2（亮 + 暗）” 的参数设计完全匹配，实现了平缓的呼吸灯光效

五、总结

本次实验让我深入掌握了STM32定时器中断与PWM输出的核心原理及PSC/ARR周期计算方法，并通过Keil软件仿真成功验证了占空比变化。在实战中，我积累了代码重定义、ST-Link烧录失败以及示波器无波形等常见报错的排查经验，深刻体会到嵌入式开发必须软硬结合、理论与调试工具并重，为后续复杂外设控制打下了坚实基础。